

HEIMDALL

User Guide

Understand the application.

Build your DCS controls.

Manage Elite Dangerous presets.

Extend the guide with each new game.

GENERAL / DCS WORLD / ELITE DANGEROUS

VYPER INDUSTRIES | DOCUMENTATION EDITION 2026.10

Current development-build guide. Actual Windows screenshots with sample data. See the scope record on the final page.

02 Find your workflow

Modular sections. Read the shared fundamentals once, then follow the chapter for your game. Every page is bookmarked; section entries below are clickable.

HEIMDALL fundamentals Pages 3-14

Setup, identity, shared layout, editions, training, appearance and recovery.

DCS World Pages 15-30

Aircraft modules, modifiers, assignments, pages, packages and deployment.

Elite Dangerous Pages 31-44

Configurations, controller linking, native assignments, presets and reference PDFs.

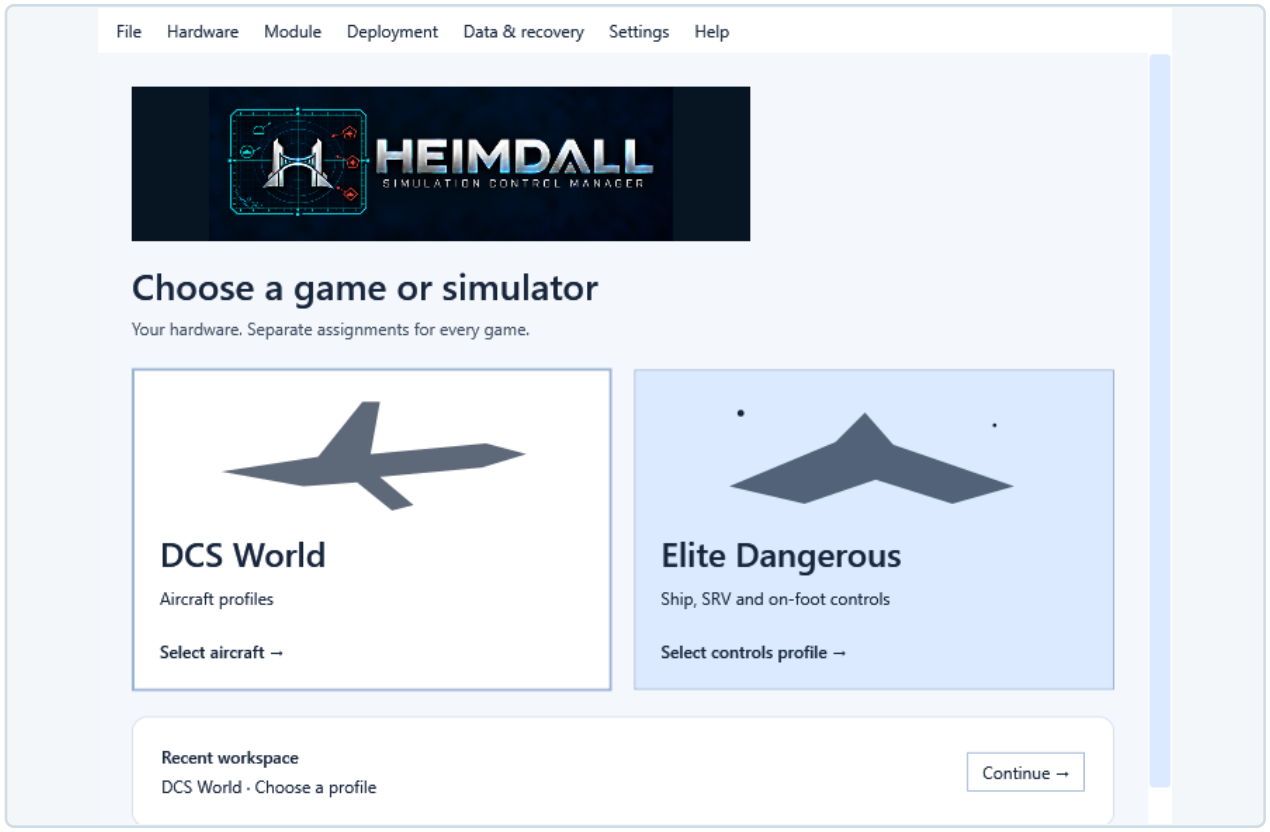
Expansion & reference Pages 45-49

Future integrations, chapter template, troubleshooting, glossary and source record.

Screenshots retain their original resolution. Zoom the PDF to inspect small UI labels. Training views are explicitly identified; sample IDs are not values to copy.

03 One application, game-specific controls

HEIMDALL is a Simulation Control Manager: organize controllers, author assignments, inspect inputs, and produce native profiles and readable references.



Actual Windows application capture - sample data.

Actual Windows application captures use synthetic test or training data. Names and device IDs are examples, not settings to copy.

1. Choose your game

DCS uses aircraft modules and module-owned UI Layer controls. ED uses named controls configurations covering Ship, SRV, On Foot and General.

2. Reuse what is shared

Hardware identity, drawings, navigation, save/discard, testing surfaces and reference rendering share a common approach.

3. Keep native meaning

Commands, modifier behavior, file formats and deployment belong to each game. HEIMDALL is not a universal cross-game input remapper.

Source detail: docs/design/game-workspaces.md

04 Choose your first route

Start from the configuration you actually have. Learn the workflow before changing a live game installation.

Your situation	Recommended route
New to HEIMDALL	Run either guided training lesson; then review hardware inventory and back up your game settings.
DCS already has bindings	Choose DCS, import that aircraft input folder, review every device and load the complete command catalog.
Starting DCS from scratch	Create a module with the exact aircraft input-folder ID and explicitly selected physical units.
Already playing ED	Import a working ED 4.x .binds preset as the native command/settings template.
Adding a new ED variant	Use New configuration or Duplicate; give each variant a distinct preset name.
Another game	A new native integration must be implemented. The final section explains the integration and documentation checklist.

Save inside HEIMDALL and exporting/deploying to a game are separate actions.

Source detail: docs/new-module-creation.md

05 Free, Premium and Owner

DCS is the only Free simulator. Existing Free hardware and appearance restrictions still apply.

Capability	Free	Premium / Owner
DCS authoring and output	Reviewed Thrustmaster catalog	All supported hardware
ED and other integrations	Unavailable	Available when implemented
Physical input test / identification	Unavailable	Available
Appearance	Fixed Navy/Amber photo preset	Skins and customization
Both DCS training lessons	Available with simulated input	Available
License	None required	Authentic signed license required

Premium and Owner differ

Premium is the customer entitlement. Owner is the application owner’s private entitlement. They currently grant the same feature access, but retain distinct identity labels.

Availability

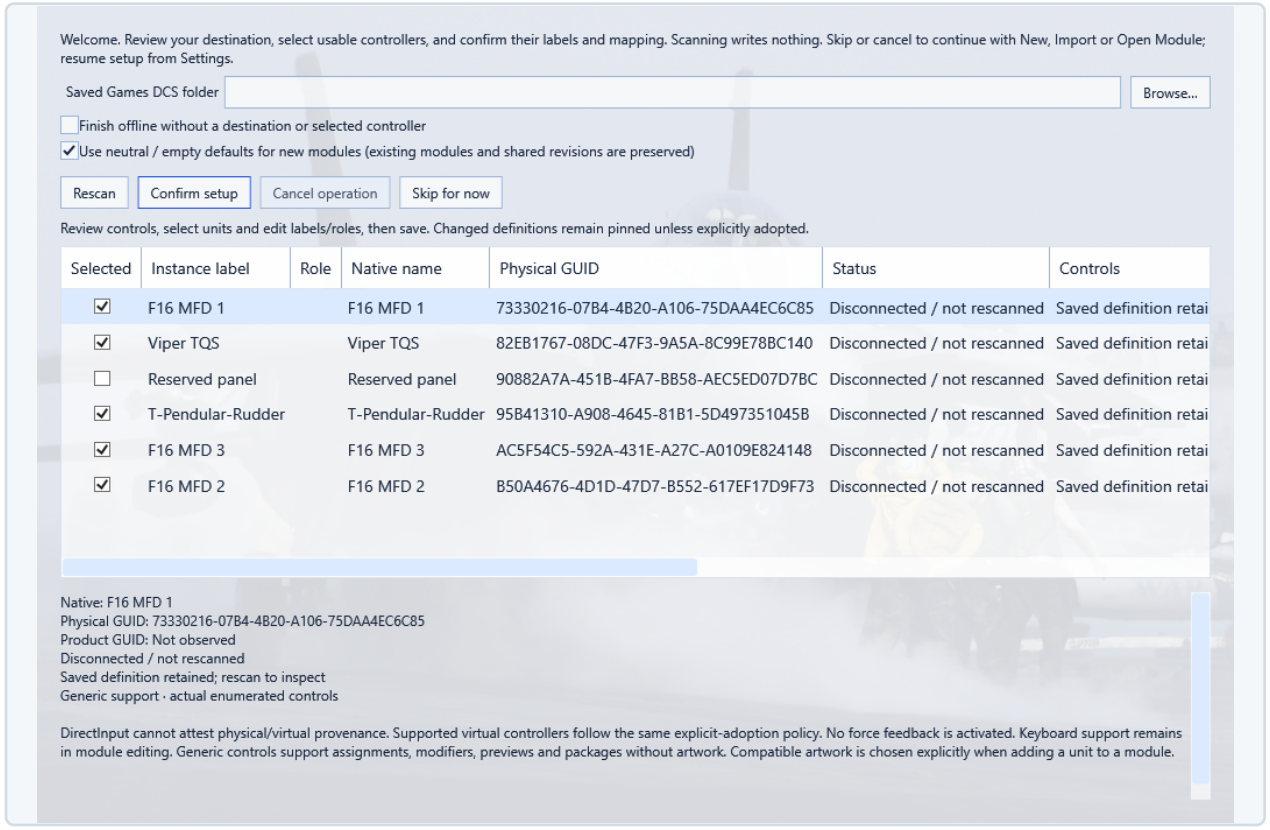
Premium pricing and commercial availability remain TBA. Offline signature acceptance does not add checkout, subscriptions, machine binding, seats or revocation.

Free is an explicit hardware allowlist, not a promise to support every Thrustmaster product. Importing a drawing never grants an entitlement.

Source detail: docs/design/free-edition.md

06 First launch and setup

Choose DCS to reach its module workflow. Hardware setup can be completed online with controllers or explicitly offline.



Actual Windows application capture - sample data.

1. Choose a destination

For DCS, browse to the Saved Games instance folder, such as C:\Users\pilot\Saved Games\DCS. Do not select its aircraft or joystick subfolder.

2. Review, then adopt

Scan results are observations. Select the physical units you want, label repeated units distinctly, and confirm only after reviewing controls and errors.

3. Return when ready

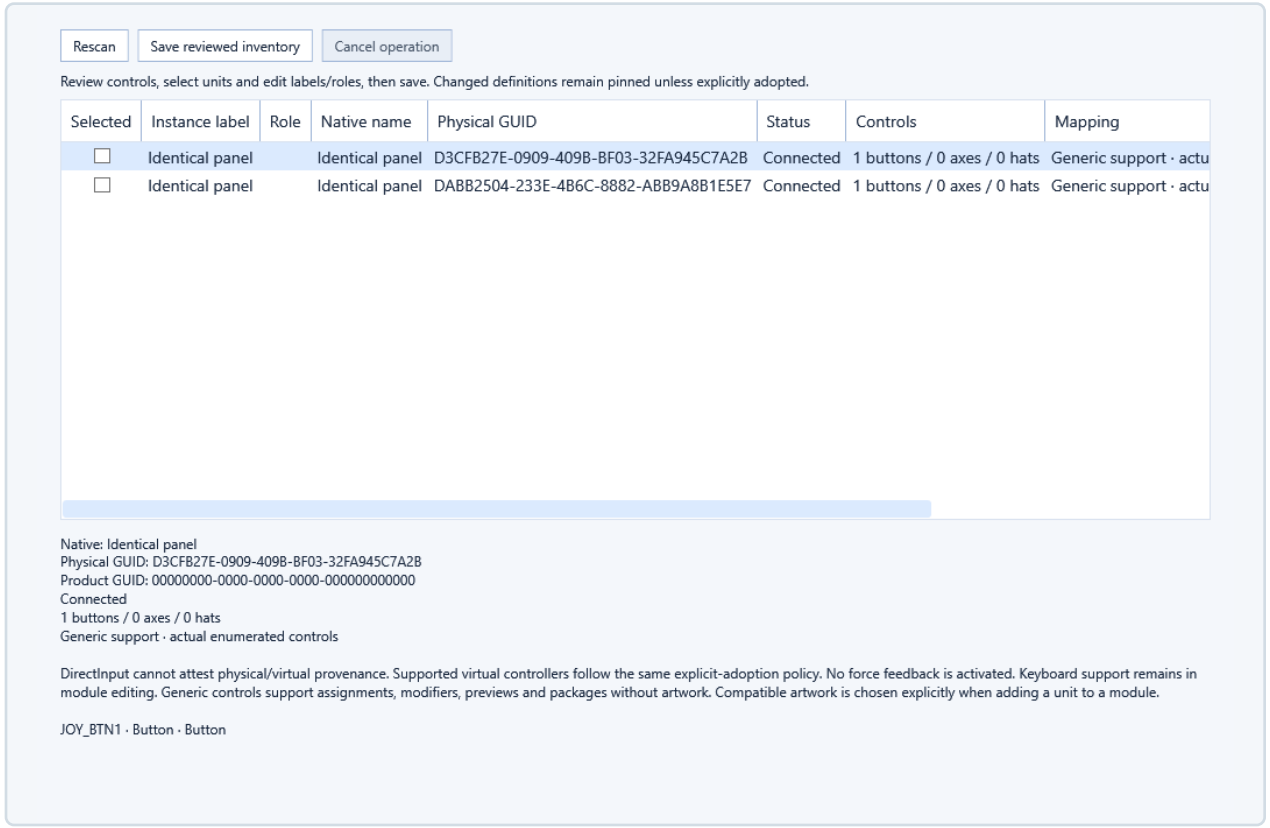
Skip for now leaves existing data intact. Resume/rerun setup is available later from Settings or profile tools.

Setup does not deploy files. A hardware template or familiar controller name does not establish which physical unit is yours.

Source detail: docs/hardware-onboarding.md

07 Hardware inventory and identity

Inventory tracks physical instances independently of module membership and artwork.



Actual Windows application capture - sample data.

Physical instance GUID = one Windows unit. Product GUID/ED product token = model evidence. Hardware template = drawing/control definition. They are not interchangeable.

1. Discover or rescan

Use Hardware > Discover devices / Rescan. Inspect native name, physical instance GUID, product GUID, control counts and diagnostics.

2. Label repeated devices

Give identical MFDs or quadrants distinct labels/roles. Select the units, then Save reviewed inventory. A scan does not add them to an aircraft.

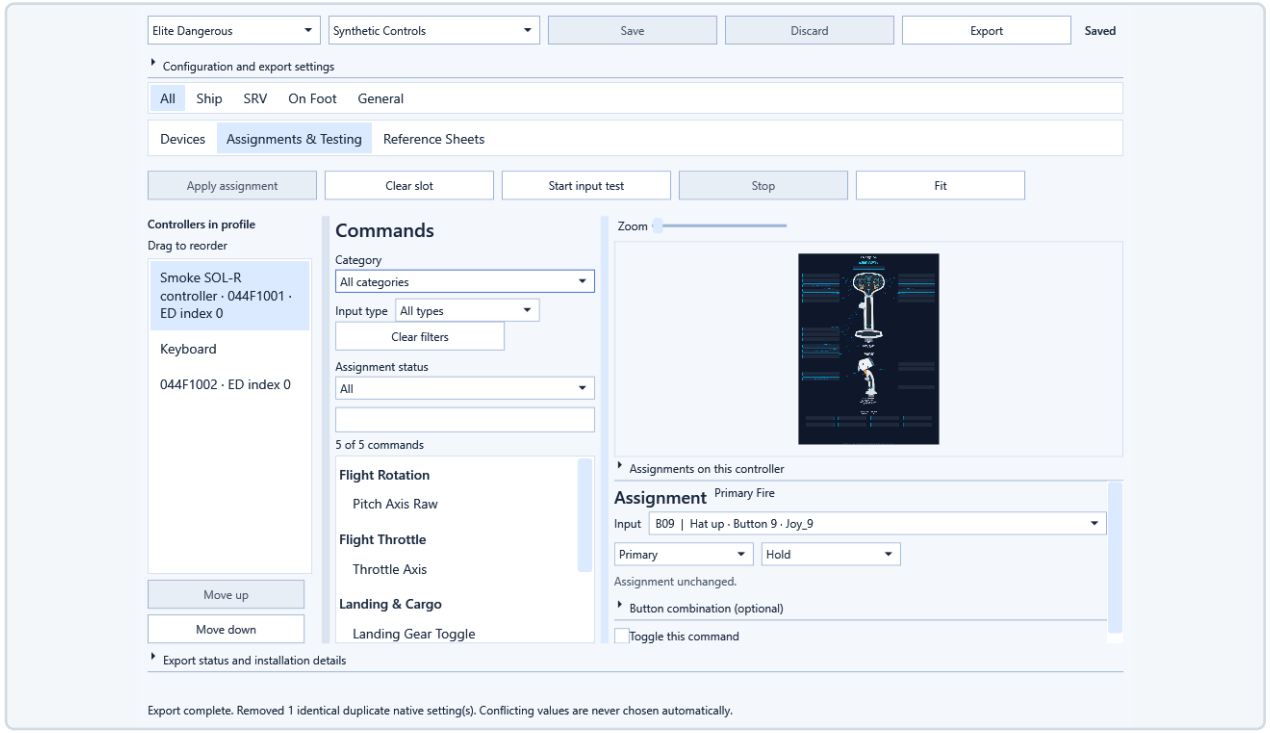
3. Review change evidence

Changed controls require explicit adoption. Offline saved units retain pinned definitions. A replacement instance requires deliberate rebinding.

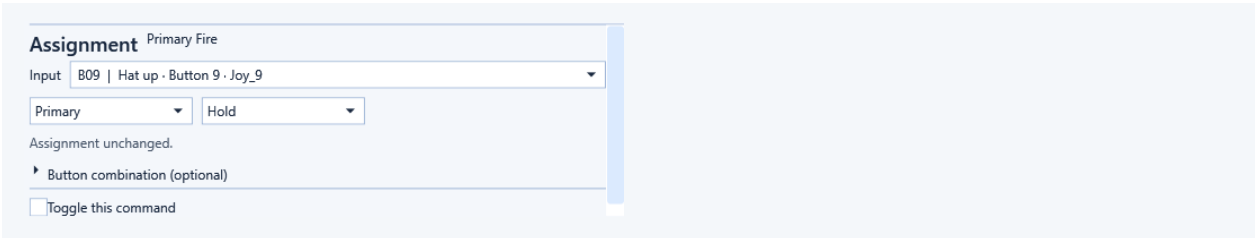
Source detail: docs/hardware-inventory.md

08 The shared workspace

Both games use Controllers, a tools/commands pane, and an assignment or reference work area.



Actual Windows application capture - sample data.



1. Keep context

Select a controller on the left. Change the feature above the work area; the controller selection follows the task.

2. Resize what matters

Drag the vertical dividers. Focus a divider and use Left/Right for keyboard resizing. Scroll an individual pane to reach longer forms.

3. Read the selected input

The inspector beneath the drawing shows the selected input and current assignment. Fit shows the complete drawing; manual zoom enables closer inspection.

Applying a binding edits the draft. The global Save/Discard controls handle the complete workspace. A drawing is optional for supported native inputs.

Source detail: docs/ui/shared-game-workspace.md

09 Apply, Save and Export

Three different boundaries protect your work. Follow the status text instead of assuming a click reached the game.

Action	What changes	What does not change
Apply / Stage	Current HEIMDALL draft	Saved workspace and live game files
Save	Persisted HEIMDALL configuration	The game’s active native preset
Discard	Restores last saved HEIMDALL state	Previously exported or deployed files
Export / Build	Produces reviewed native output or a package	The running game is not automatically configured
DCS Fly Now	Copies verified saved DCS payload	DCS is not started automatically

Work in small steps

Choose controller, scope/slot, command and native input. Apply or stage, inspect, then Save.

Before trusting a result

Load the result in the game and verify the intended physical controls.

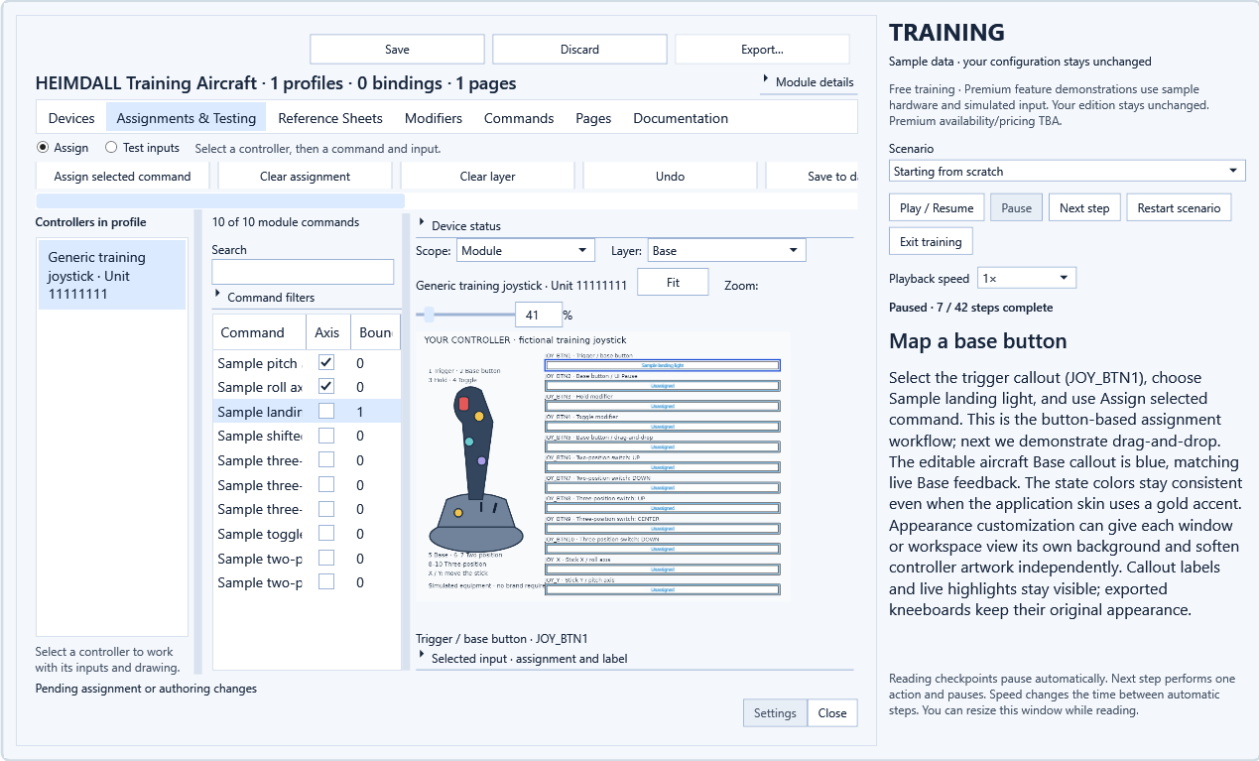
Before output

Save or discard pending edits. Use the correct native destination and review replacement warnings.

Source detail: docs/module-authoring.md

10 Learn with guided training

Help > Training / Guided workflows opens a separate demonstration session. Both lessons are Free.



Training demonstration - sample data.

1. Choose a scenario

Starting from scratch demonstrates creation. Starting with a preconfigured module in DCS demonstrates import and refinement.

2. Control playback

Play/Resume runs the demonstration. Next step performs one action and pauses. Reading checkpoints pause automatically; speed changes the interval between actions.

3. Restart or exit safely

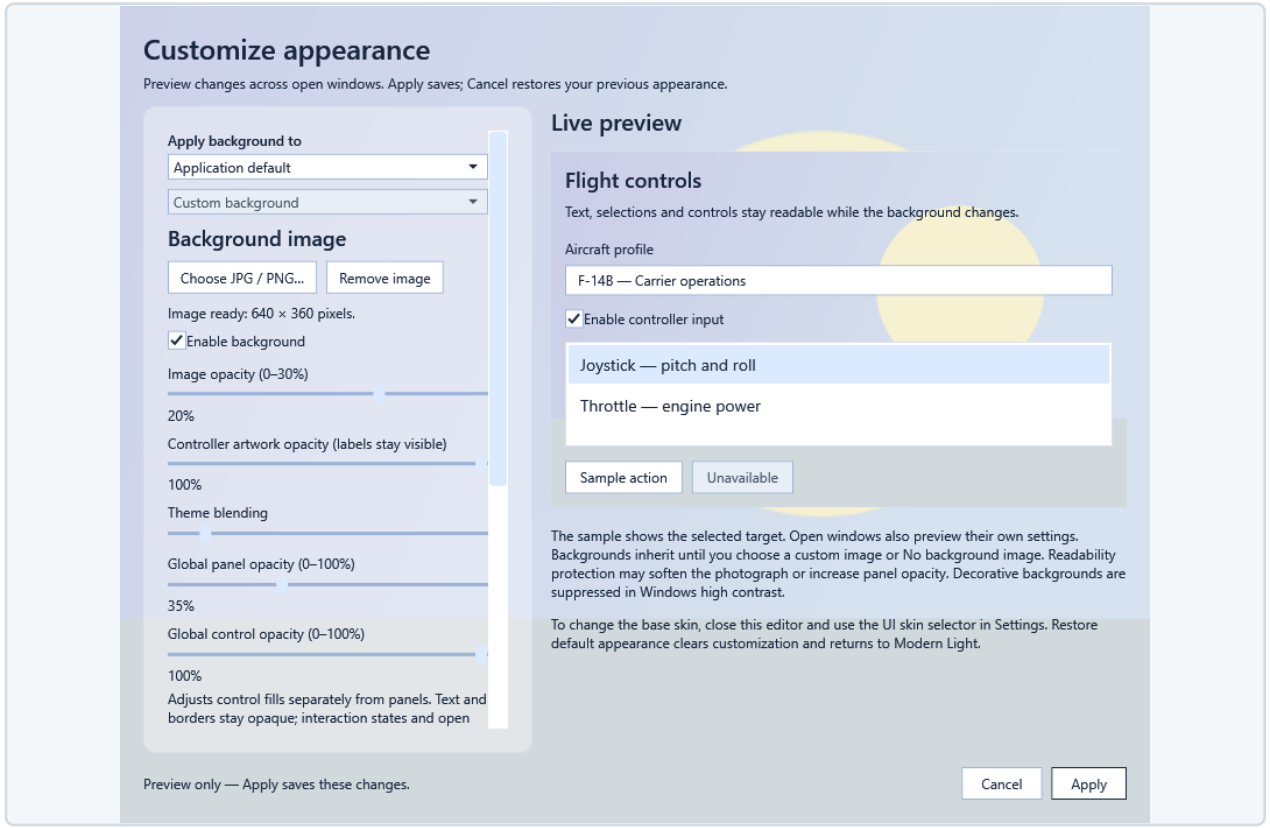
Restart creates fresh sample data. Exit settles active work and removes training files. Normal modules, settings and Saved Games remain separate.

Training uses a fictional joystick and simulated input. It demonstrates Premium capabilities without unlocking physical testing. Sample command codes and GUIDs are not an aircraft profile to install.

Source detail: docs/interactive-training.md

11 Appearance without changing output

Premium/Owner can select Light, Dark, Navy/Amber or Cockpit Green and customize application surfaces.



Actual Windows application capture - sample data.

1. Preview first

Settings > Customize appearance provides image, fit/alignment, blend, panel/control opacity and color previews. Apply commits; Cancel restores accepted settings.

2. Target a surface

Choose application default, a window/dialog, or an individual workspace. Use inherited background, custom background, or no image.

3. Separate preview from output

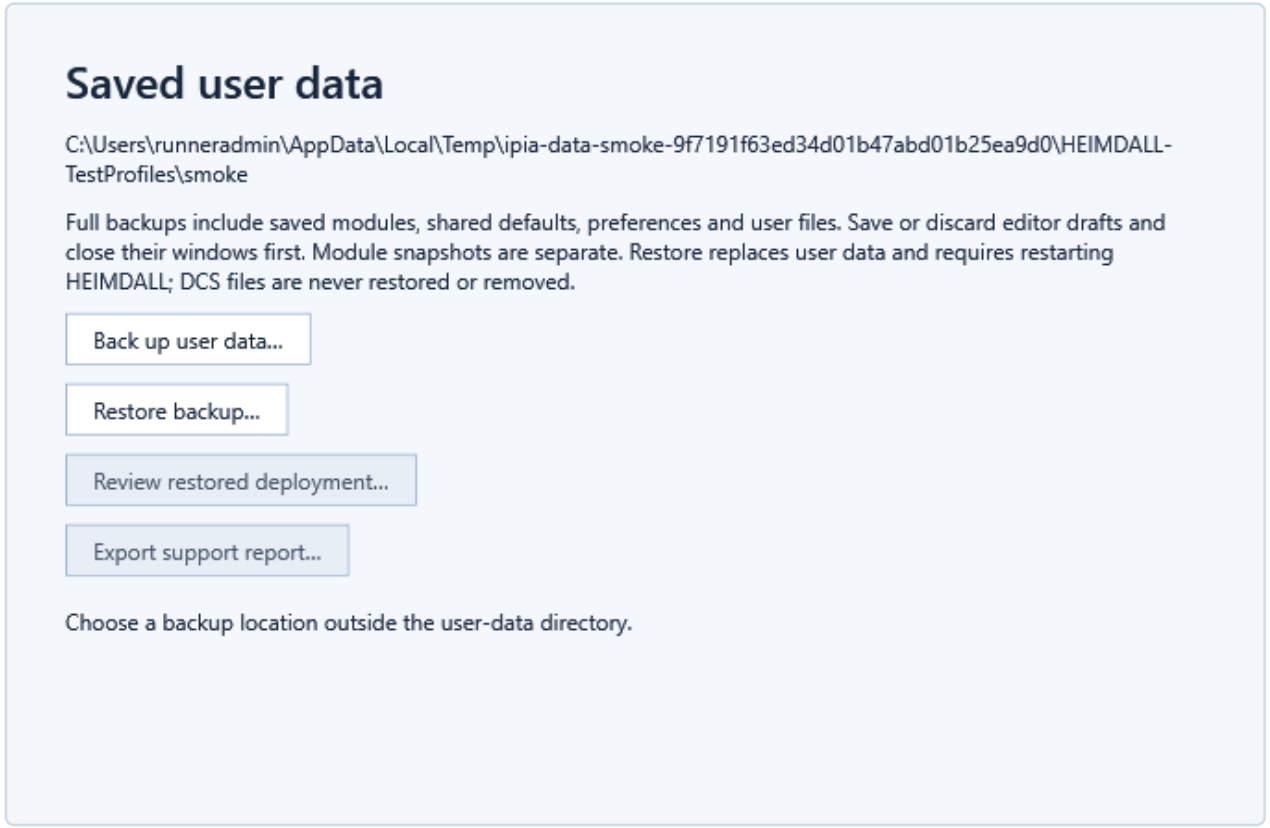
Controller artwork opacity changes preview decoration. Callout text and live highlight frames remain opaque; packaged kneeboards keep their original output appearance.

Free retains its fixed preset and saved paid customization. Readability protection or Windows high contrast can suppress decoration or increase opacity.

Source detail: docs/design/appearance-editor.md

12 Back up the whole working profile

Use Data & recovery for a verified full-profile archive. Save or discard open drafts first.



Actual Windows application capture - sample data.

A full backup restores a profile. A DCS module snapshot restores one module; an ED .binds export is native game output. These are different recovery tools.

1. Create a verified backup

Choose a new .heimdall-backup.zip outside the active data directory. The archive includes the database, profile settings, user assets and recovery data.

2. Understand the boundary

Installed binaries, your DCS Saved Games files and the account-wide signed license are not included. Back those up independently when needed.

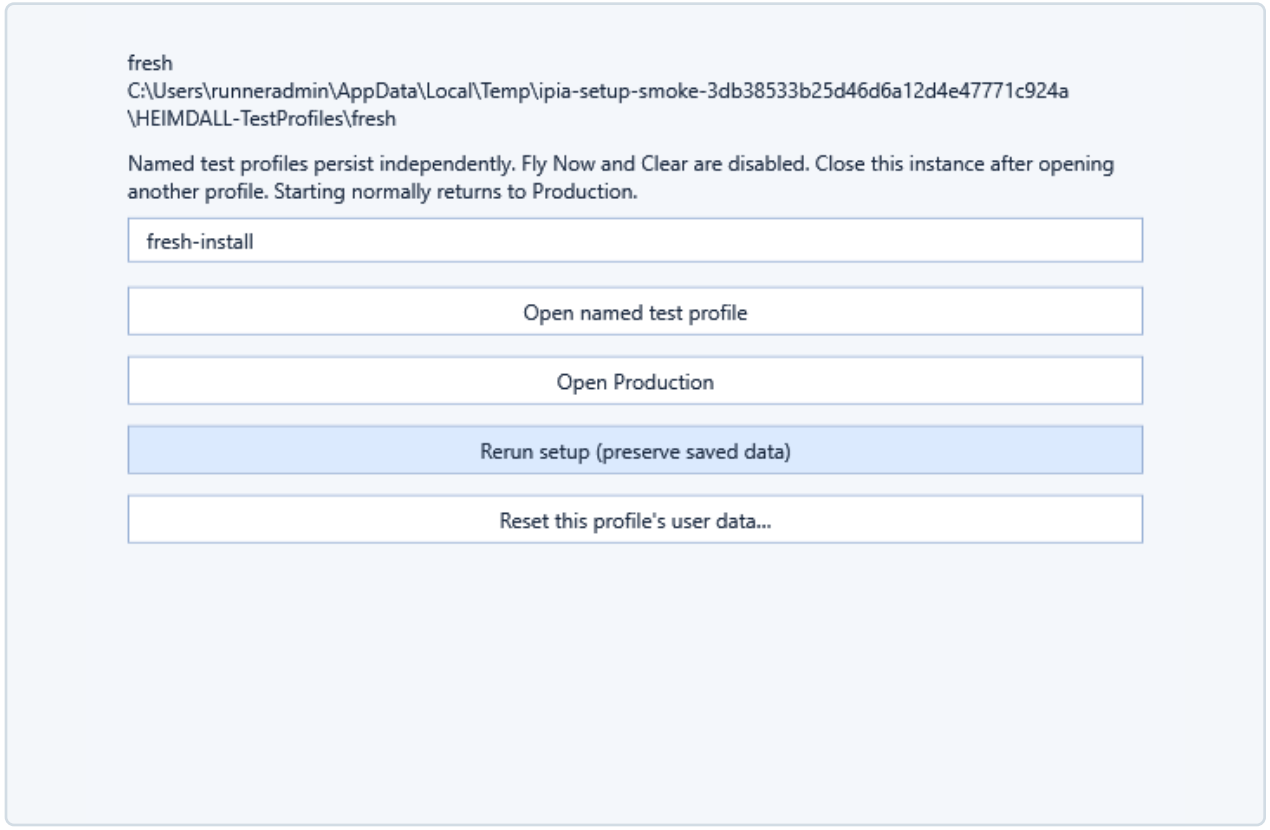
3. Restore deliberately

Restore verifies data before replacement, retains the previous directory and requires restart. Restored deployment records require explicit review before Fly Now or Clear.

Source detail: docs/operations/database.md

13 Isolate experiments and test editions

A named test profile isolates working data; licensing remains Windows-user-wide.



Actual Windows application capture - sample data.

Fly Now and Clear are disabled in named test profiles. Use a normal installation inside a disposable VM for full deployment acceptance.

1. Open an isolated profile

Use Data & recovery > Test profiles and reset, or launch Heimdall.exe --test-profile practice. New profiles have independent modules, settings, inventory and drawing files.

2. Know the shared settings

Workspace layout and the signed license are shared for that Windows account. A new profile name does not turn Owner into Free.

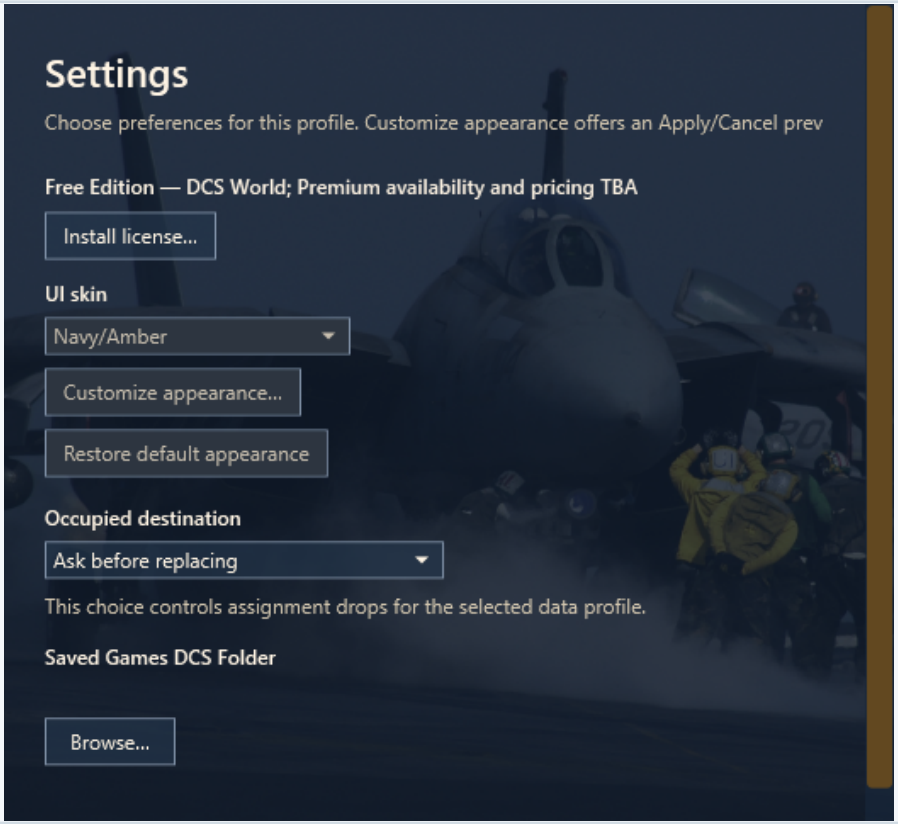
3. Test in a VM

Use separate Windows accounts or private VM checkpoints for Free, Premium and Owner. Confirm real native controller access before treating VM tests as hardware acceptance.

Source detail: docs/operations/test-profiles.md

14 Activate or return to Free

Settings > Install license accepts a signed Premium or Owner JSON document. Restart to apply it.



Actual Windows application capture - sample data.

1. Install an authentic license

Choose the privately issued JSON file. Invalid imports do not replace an accepted license. The displayed edition identifies Free, Premium or Owner.

2. Reversibly test Free

Close all HEIMDALL instances. Preserve your working license; rename owner-license.json to owner-license.json.disabled in both current and legacy licensing folders if present.

3. Restore paid access

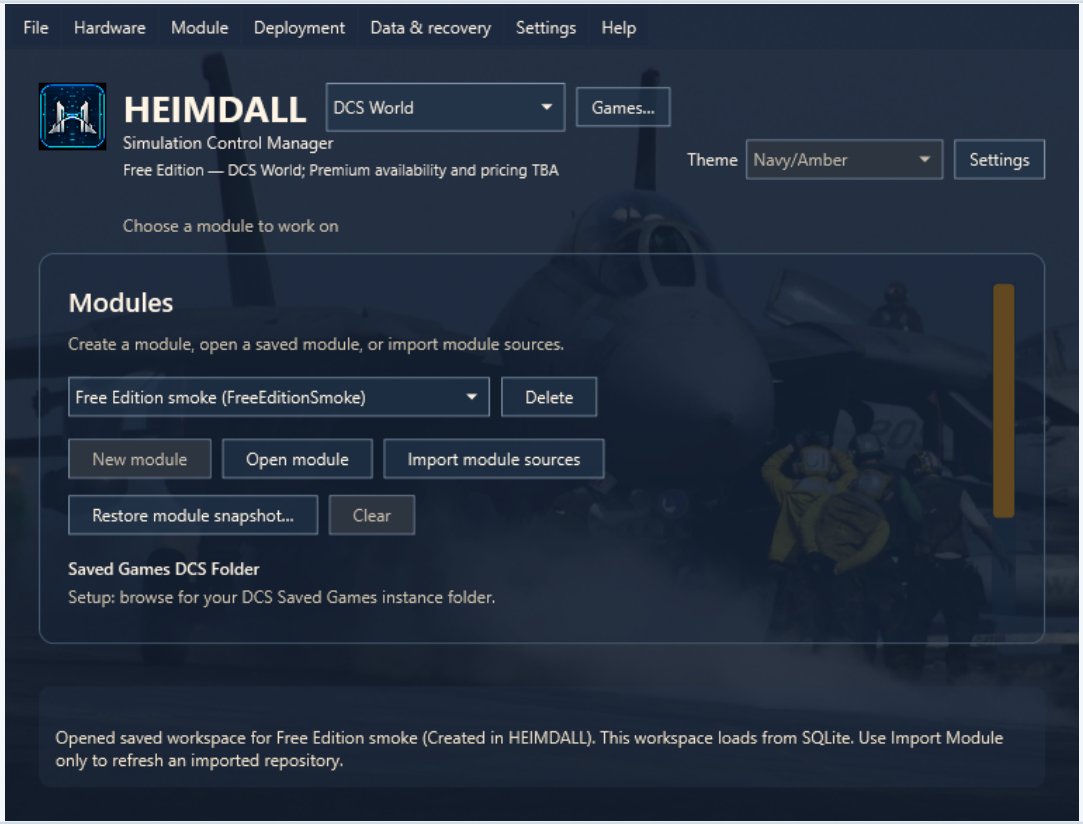
Restore the original filename, or install the preserved valid JSON through Settings, then restart. Profiles and appearance preferences remain intact.

Current: %LOCALAPPDATA%\VyperIndustries\HEIMDALL-Licensing. Legacy fallback: %LOCALAPPDATA%\VyperIndustries\DCS-IPIA-Licensing. Keep signed licenses and issuer keys private.

Source detail: docs/design/free-edition.md

15 DCS: aircraft modules and UI Layer

An aircraft module holds its native DCS identity, devices, assignments, commands, pages and retained source evidence.



Actual Windows application capture - sample data.

1. Select the aircraft

Choose DCS, then create, import, restore or open a module. The display name and exact DCS input-folder identity have different jobs.

2. Select the scope

Module means aircraft controls. UI Layer means this module's copy/overlay of global DCS functions. Each scope has its own assignments and modifiers.

3. Keep shared defaults deliberate

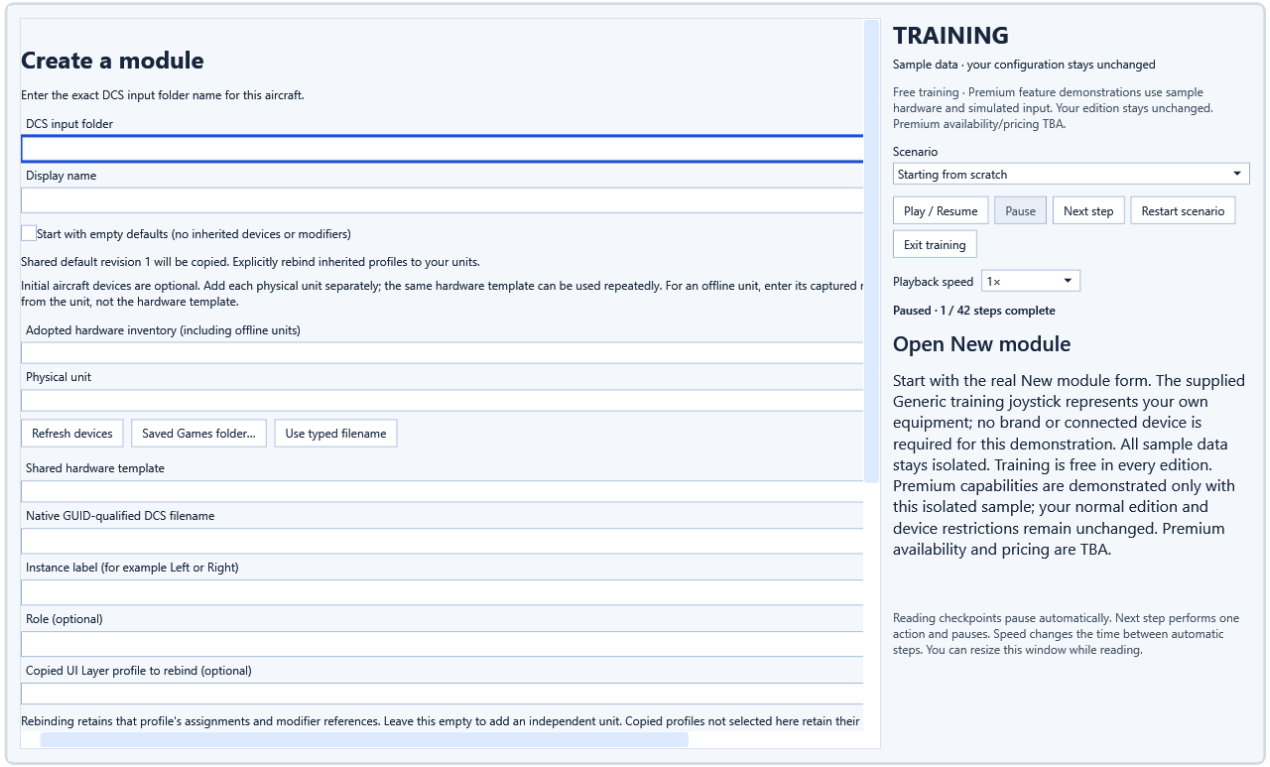
An explicitly copied shared revision is pinned. Editing one module does not silently update every aircraft or the independent shared-default workspace.

The Modules screen owns saved-module selection and deployment. The workspace owns authoring. Output actions operate on saved state.

Source detail: docs/design/module-workspace.md

16 Create or import the aircraft

Use the aircraft’s real DCS input identity and actual GUID-qualified native filenames.



Training demonstration - sample data.

Shown: the actual creation form inside training. Native diff files omit unchanged DCS defaults. Importing the folder does not invent a complete aircraft command catalog.

1. Create from scratch

New module asks for DCS input-folder ID and display name. Start with empty defaults unless you intentionally need a copied shared revision.

2. Add explicit units

Choose a physical unit and compatible hardware template, review its native filename, then add it. Repeated units require separate entries.

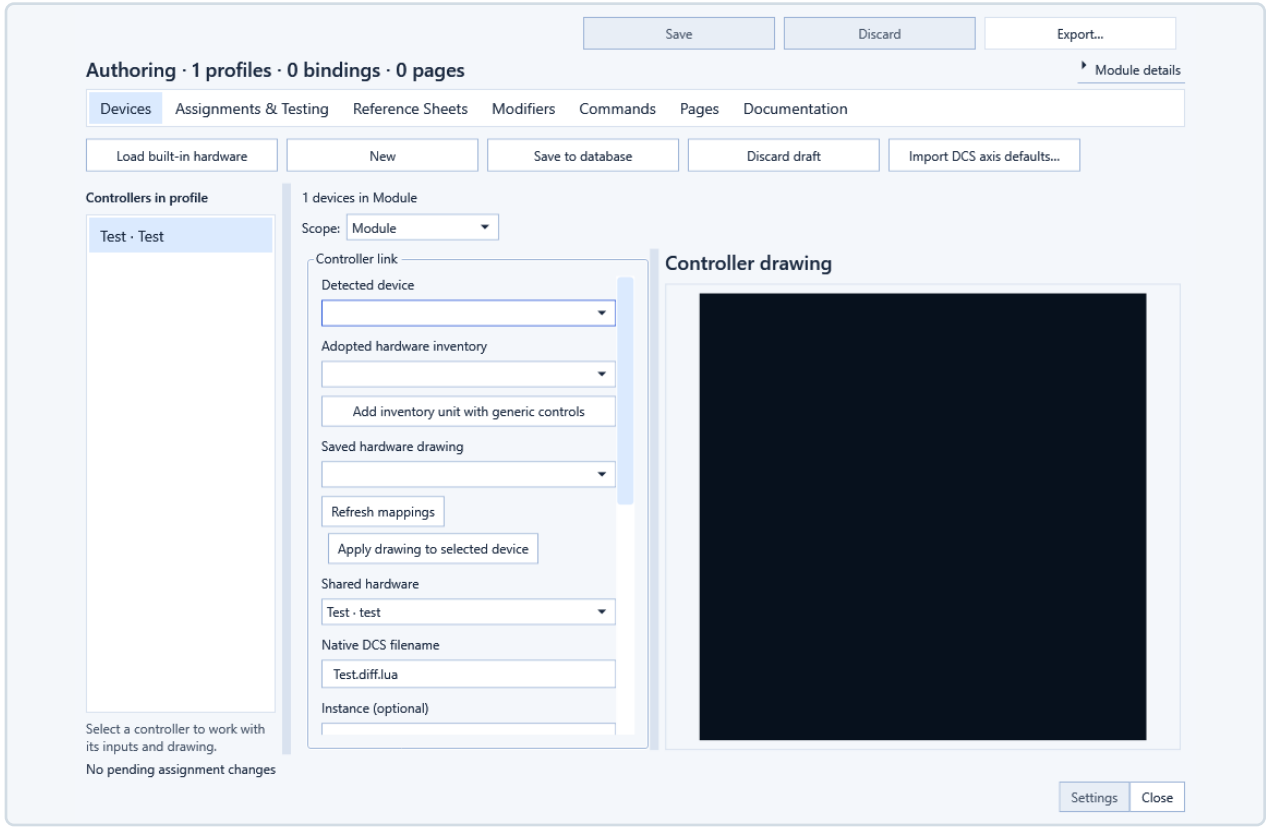
3. Import existing controls

Import module sources reads an aircraft input folder or joystick subfolder. Review every device, including empty/removal-only profiles, before confirming.

Source detail: docs/new-module-creation.md

17 Devices: identity, scope and artwork

Devices controls module membership and native identity; an illustration does not choose the physical hardware.



Actual Windows application capture - sample data.

Removing the final aircraft/UI membership removes that device’s combined pages and assignments. A profile retained in the other scope stays represented.

1. Review the selected row

Check instance label, hardware identity, scope and exact native filename. Confirm the physical unit belongs to you before building.

2. Rebind explicitly

Use Rebind physical unit to change an existing unit’s GUID-qualified filename while preserving authored state and modifier references. Review the old/new identities.

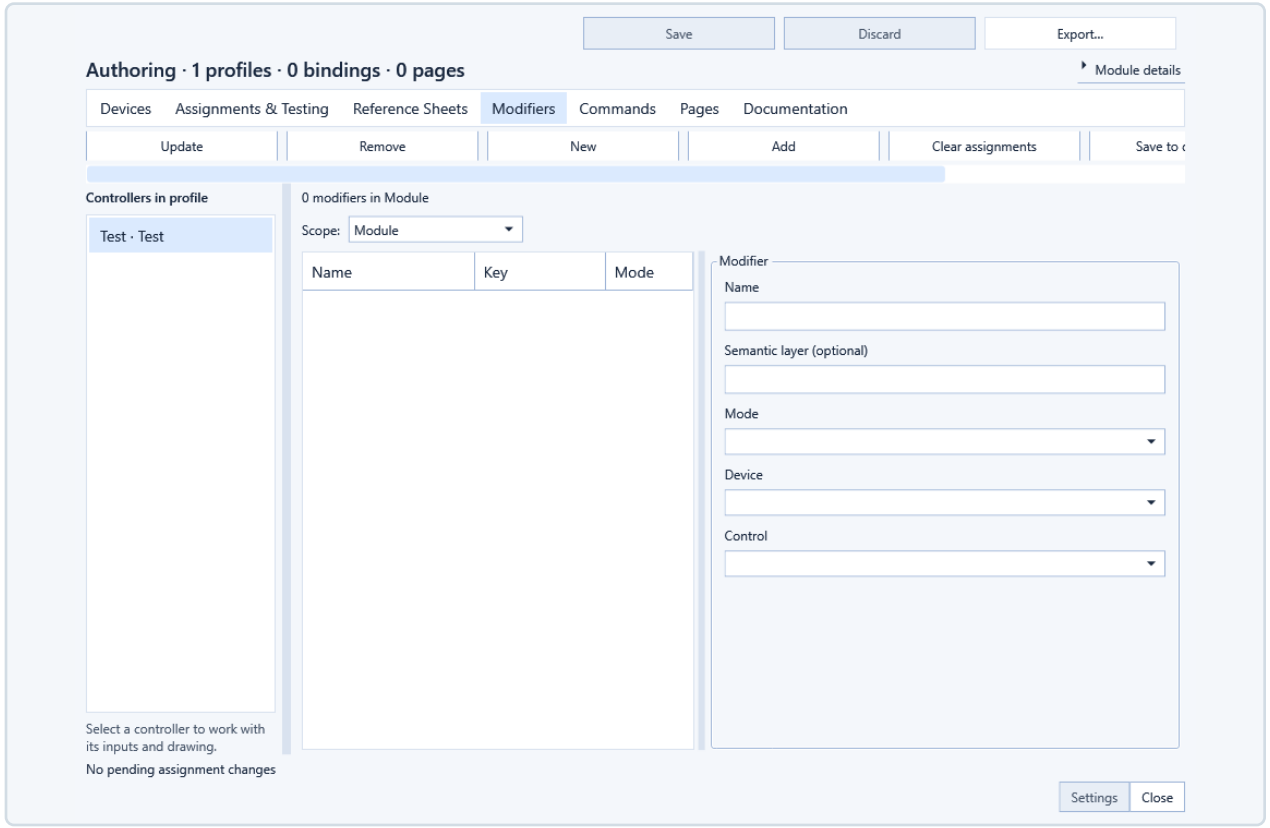
3. Stage structural changes

Add/remove scope membership, change labels/headings, select a drawing, or add adopted generic controls. Save keeps the draft; Discard restores saved state.

Source detail: docs/new-module-creation.md

18 Modifiers: Hold, Toggle and chords

DCS modifiers are named activators in the selected scope. They can live on another controller.



Actual Windows application capture - sample data.

Hold is active while pressed. Toggle changes on a new press; releasing it does not clear the latch. Test both press/release behavior and scope overlap.

1. Define an activator

Select its physical device and native input, give it a useful name, and choose Hold or Toggle as appropriate.

2. Choose the correct layer

Base has no modifier. A shifted/chord layer applies only when its defined activators are satisfied. Aircraft and UI Layer definitions remain separate.

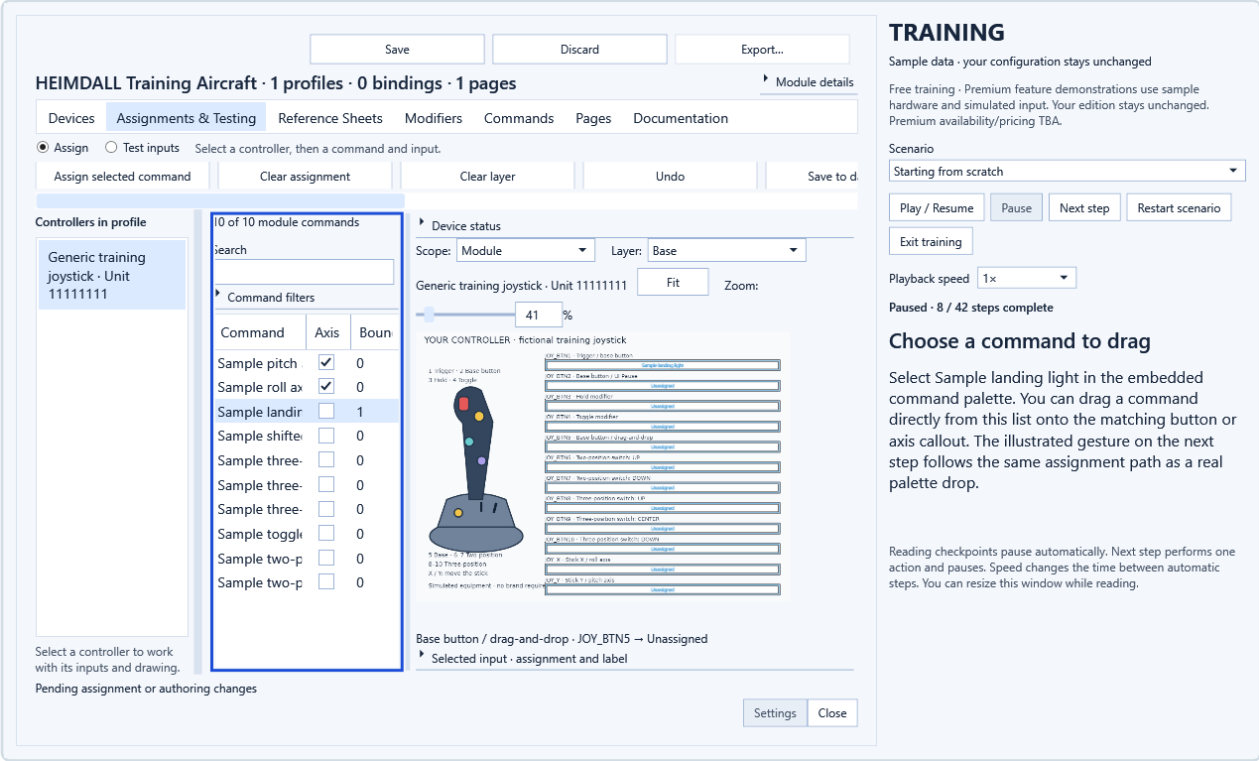
3. Respect reserved controls

An input serving as a modifier activator cannot also receive an ordinary assignment until the modifier is changed or removed.

Source detail: docs/live-input-testing.md

19 Assignments: command to physical input

Choose controller, scope and layer before selecting the command and control.



Training demonstration - sample data.

Shown: real Assignments inside the isolated training session. Controller selection is preserved when switching between Devices, Assign and Test inputs.

1. Choose a command

Search/filter the aircraft or UI Layer command palette. Button and axis commands need the corresponding kind of input.

2. Assign two ways

Select a callout and use Assign selected command, or drag a compatible command onto the intended control. Inspect the resulting selected-input readout.

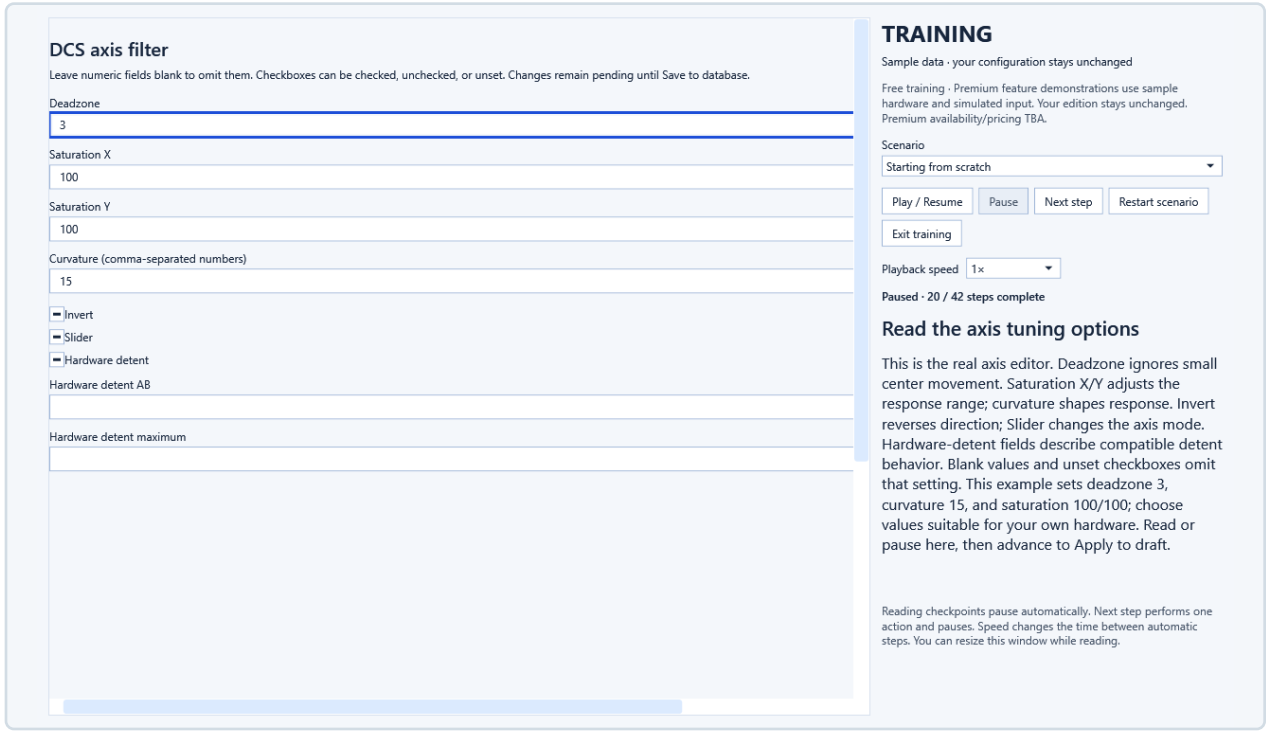
3. Keep the draft deliberate

Save persists the module’s staged work. Discard restores the saved workspace. Importing command definitions does not apply their captured assignments.

Source detail: docs/module-authoring.md

20 Labels, context actions and axis tuning

Right-click a callout to change its displayed label, clear its assignment or tune an axis.



Training demonstration - sample data.



1. Distinguish label from identity

A displayed assignment label changes the caption for that scope/layer. A generic control name describes the physical input. Neither changes its native key.

2. Tune the exported filter

The DCS axis editor supports deadzone, saturation X/Y, curvature, invert, slider and hardware-detent fields. Blank numeric fields omit that value; checkboxes support unset.

3. Apply, then save

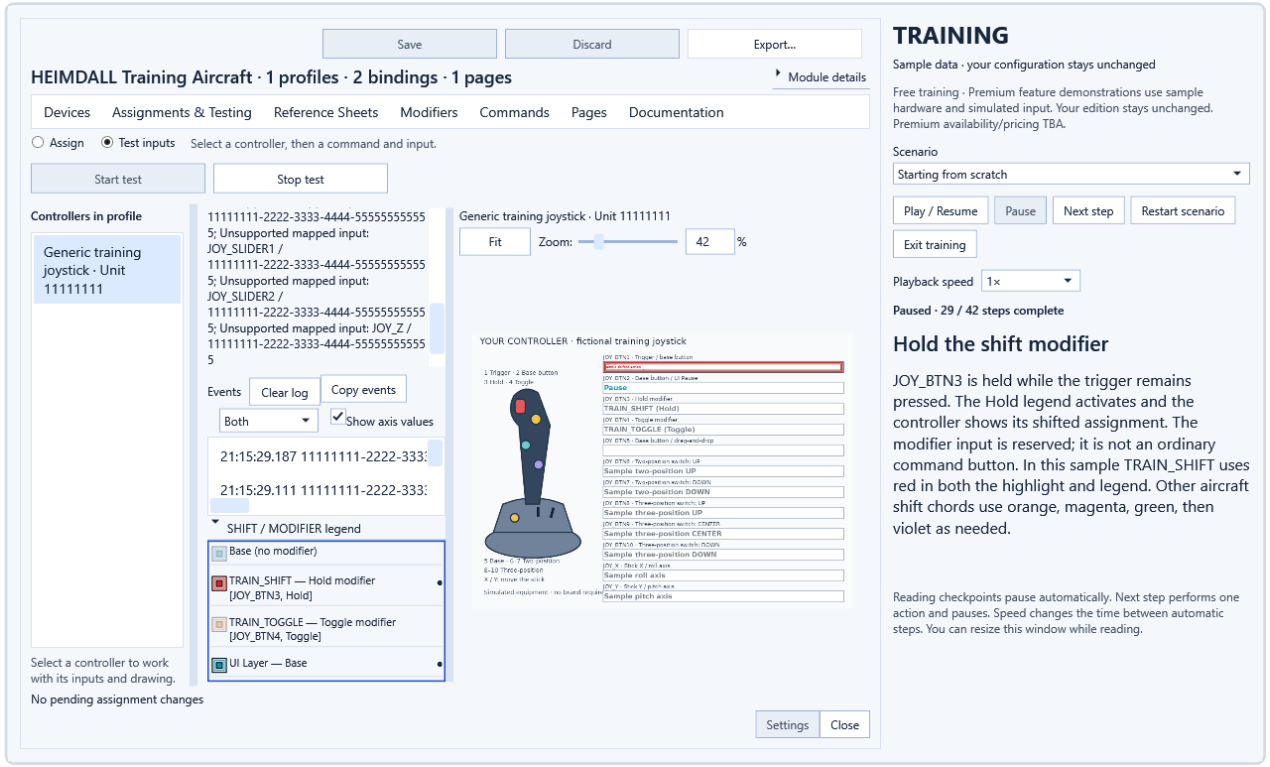
Apply to draft stages tuning. Save to database persists it. Clear tuning removes the explicit filter values.

Search & replace labels supports a live review, literal case-insensitive matching, and optional regex. Review scope and affected controls before Apply.

Source detail: docs/module-authoring.md

21 Test current edits or saved state

Choose Test inputs under Assignments & Testing. Physical testing requires Premium/Owner.



Training demonstration - sample data.

1. Freeze the intended version

Choose Current edits or Saved profile, then Start test. Stop/Start after edits to generate a fresh test snapshot.

2. Operate one selected unit

Feedback is filtered by exact physical GUID. The log shows native input, axis value, active modifiers and matching command identities.

3. Read scope and modifier feedback

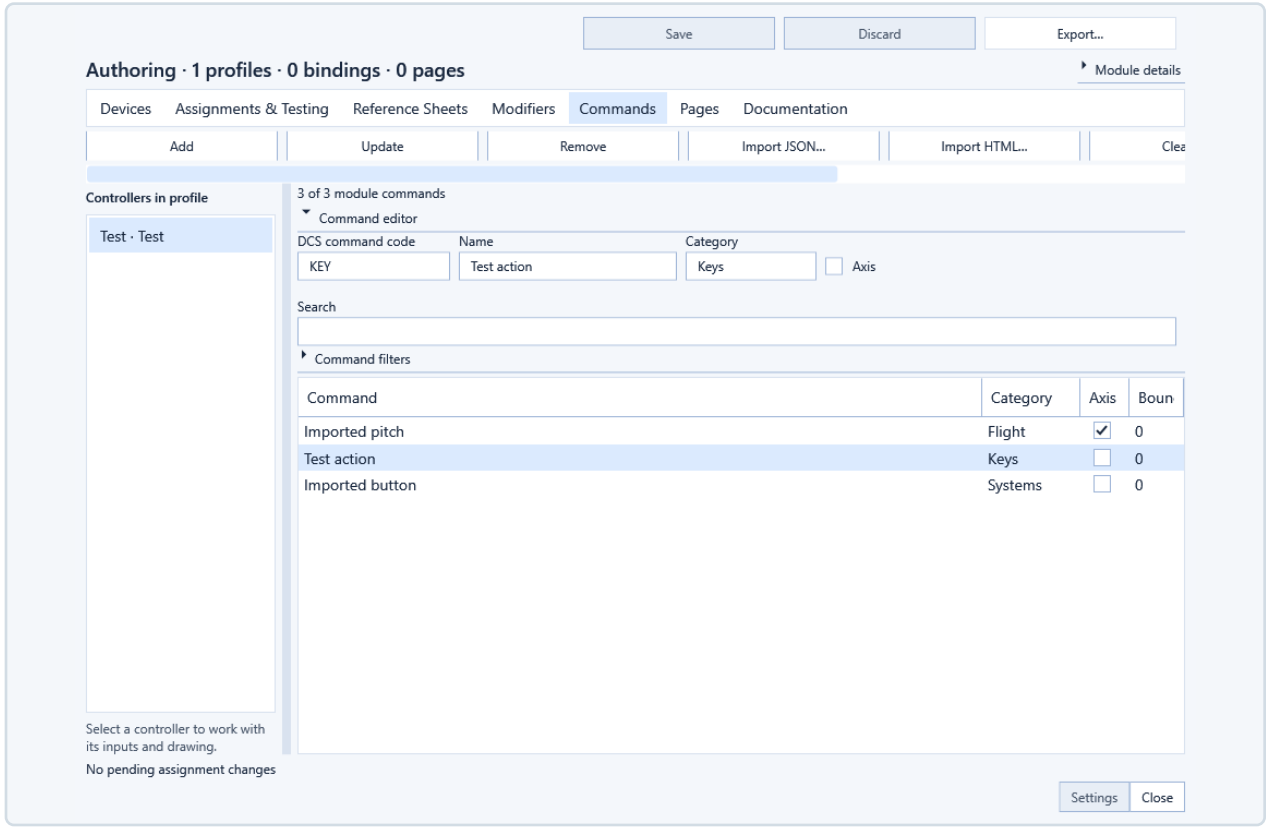
Aircraft Base uses blue; UI Base uses cyan. The legend identifies chord colors. Focus loss, device loss and Stop release input handles.

Training shows simulated feedback. Physical tests send no game or force-feedback commands. Raw axis values and exported metadata are not an exact DCS curve simulation or proof of cockpit behavior.

Source detail: docs/live-input-testing.md

22 Commands: build the aircraft catalog

Commands defines available native actions. Display names and categories do not replace command codes.



Actual Windows application capture - sample data.

The palette initially includes only commands evidenced by imported diff files. Load the full aircraft catalog to expose unassigned commands and cover axis cleanup.

1. Import accurate definitions

Load IPI JSON for the same module ID, or DCS Controls Generate HTML exports. Review the destination aircraft and proposed additions/updates.

2. Author where appropriate

Add exact DCS command code, readable name, category and key/axis type. Command identity remains fixed after creation.

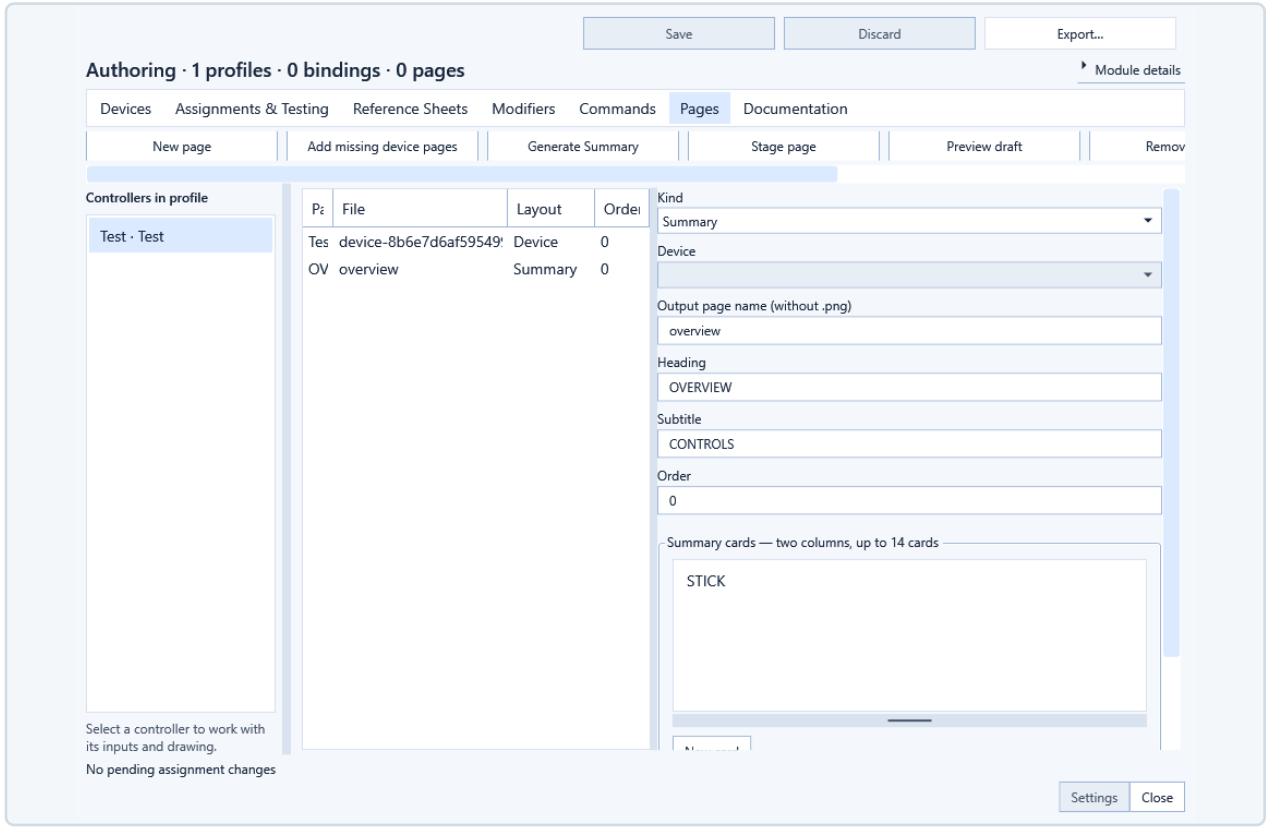
3. Preserve bindings

Assigned commands cannot be deleted or changed between key/axis until assignments are cleared. Imports retain unrelated commands, bindings, tuning and labels.

Source detail: docs/module-authoring.md

23 Pages: device cards and summaries

Pages authors the reference output; viewing a drawing is not the same as defining its package pages.



Actual Windows application capture - sample data.

1. Add device pages

Add missing device pages creates combined pages for pinned controllers. Set safe output basename, title, subtitle and order.

2. Create an overview

Generate Summary appends an overview of current device pages. Summary cards support headings, descriptions and accents; long content expands and paginates.

3. Preview and stage

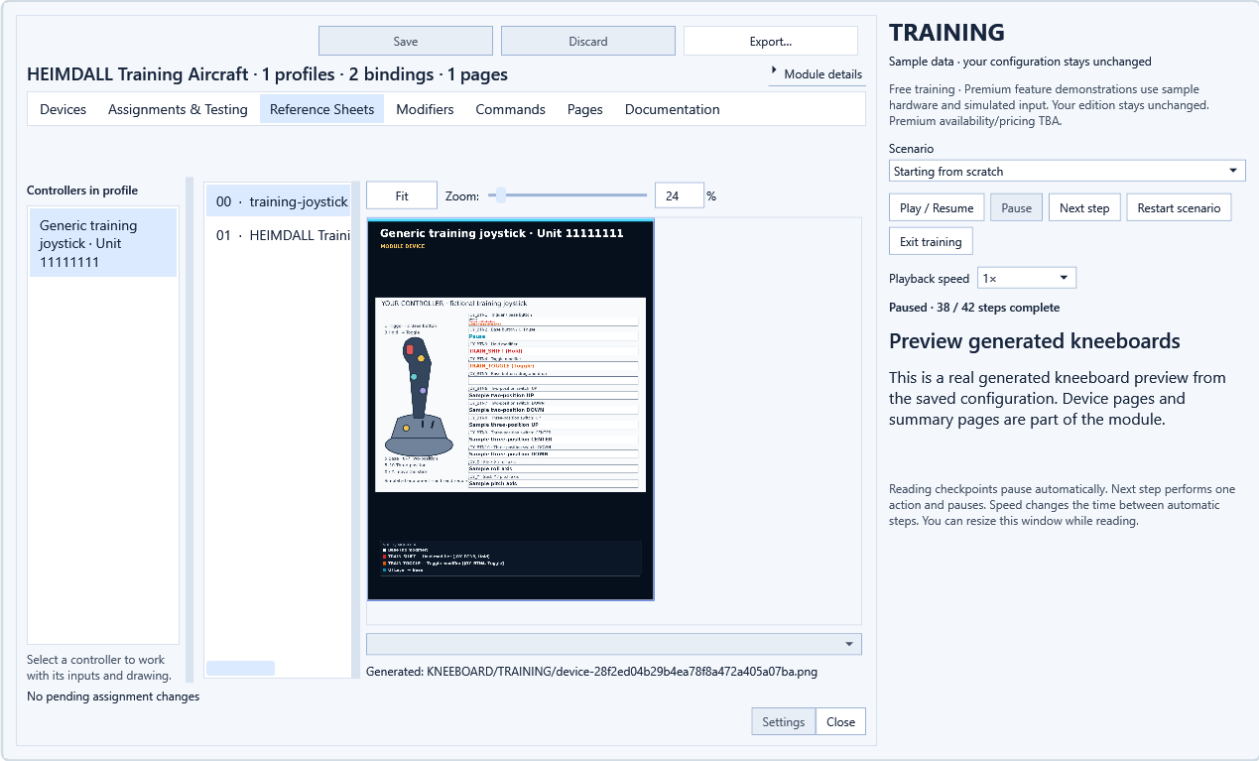
Preview draft renders current form values. Stage page validates and stages them. Reorder cards by dragging, then stage and Save.

Removing a page does not remove its device. Generate another summary after page changes to refresh overview references; existing summaries are not silently rewritten.

Source detail: docs/module-authoring.md

24 Reference Sheets: saved kneeboards

DCS Reference Sheets uses the native kneeboard page workflow. The stable feature key remains Kneeboards.



Training demonstration - sample data.

Training provides a populated example of a joystick kneeboard. Exported/packaged pages remain opaque and keep their output colors regardless of app background customization.

1. Choose the generated page

Select a device or summary page. Fit shows its complete bounds; manual zoom keeps the chosen scale while resizing.

2. Know the source state

This preview renders saved state. Draft page authoring and generic draft previews are available in their editing flows.

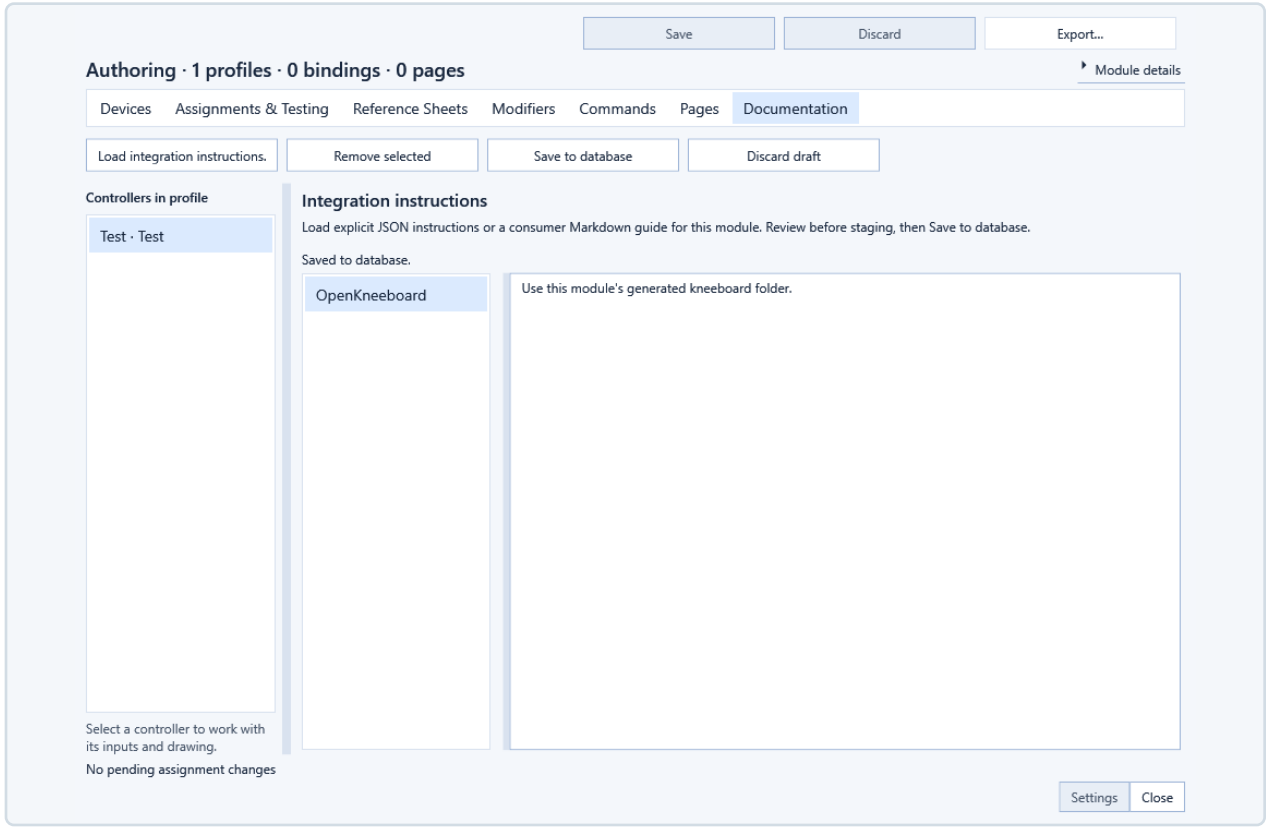
3. Check dense controls

Review headings, assigned text, modifier descriptions and continuation pages before packaging. Generic controllers can span several readable pages.

Source detail: docs/generic-hardware.md

25 Documentation: operator instructions

Documentation retains explicit integration instructions with the module. It does not configure another application.



Actual Windows application capture - sample data.

1. Load reviewed text

Choose Load integration instructions and select supported JSON or a consumer Markdown guide. Review the destination and old/proposed text.

2. Stage and save

Stage import changes the draft. Save persists it; Discard restores saved groups. Remove selected deletes the effective instruction group.

3. Export with the module

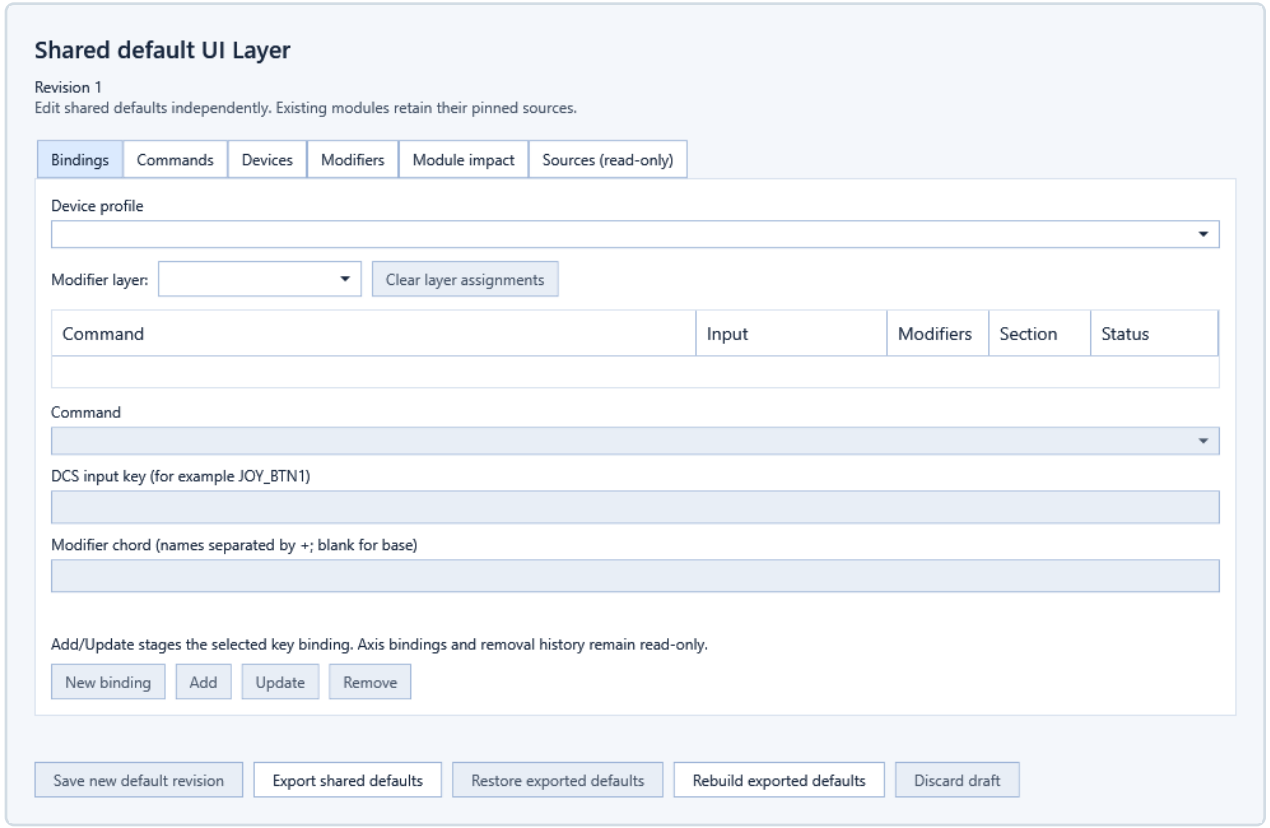
Saved groups appear in generated INTEGRATIONS.md and Complete release documentation. Captured historical paths need review against the generated installation guide.

VAICOM, OpenKneeboard and other integrations are not invented or enabled by importing text. Scripts, credentials and integration executables are not deployed by this feature.

Source detail: docs/module-authoring.md

26 Shared UI Layer: baseline versus module

Shared defaults provide a reviewed starting revision; module-owned UI controls remain independent.



Actual Windows application capture - sample data.

The capture shows the shared workspace’s retained source preview. Global defaults, module-owned overlays and the UI Layer currently installed into DCS are separate states.

1. Edit the shared workspace deliberately

Maintain reusable global command definitions, devices, modifiers and bindings in the shared UI Layer workspace.

2. Choose an explicit baseline

A new module can start empty or pin a selected shared revision. Review/rebind copied physical units before packaging.

3. Adopt changes explicitly

Changing shared defaults does not rewrite existing modules. A module must review/adopt another revision through its guarded workflow.

Source detail: docs/migration/shared-default-workspace.md

27 Build & export: choose the right artifact

Save or discard pending edits, choose a package version, then validate the saved module before distributing output.

Output	Purpose	Install directly?
.heimdall-module.json	Portable complete authoring/build snapshot with digest	No; restore into HEIMDALL
OvGME ZIP + checksum	Deployable DCS controls and kneeboards	Enable through OvGME
Documentation folder	Versioned installation, mappings, sources and instructions	No
Complete distribution ZIP	Contains OVGME, Documentation, Snapshots and checksums	Extract; install only nested OvGME ZIP

Validate the saved module

Validation checks identities, commands, modifiers, pinned assets, required pages and generated output.

Keep matching evidence

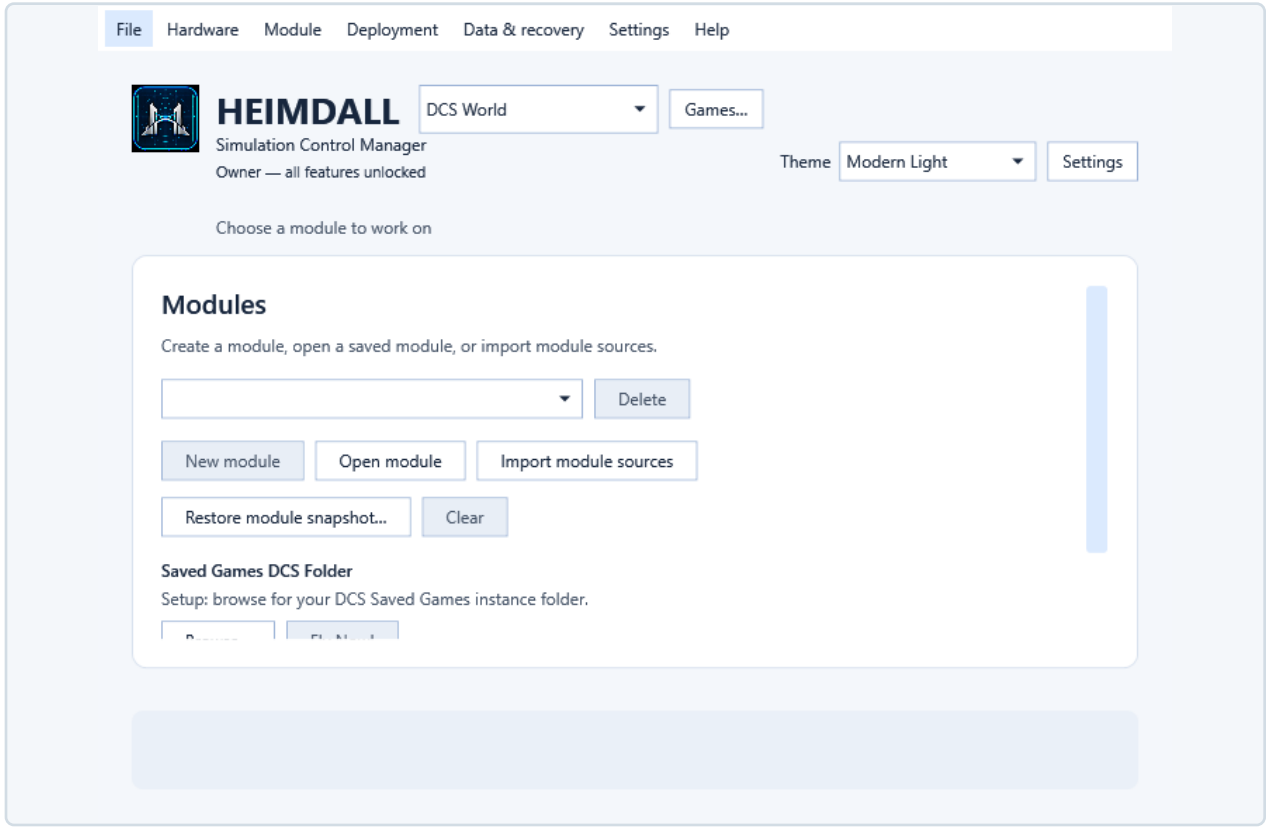
Store the snapshot, package and checksum together. A restored snapshot can reproduce its frozen build without rereading your original sources.

The Complete ZIP is not itself an OvGME mod. A module snapshot is not a full-profile backup.

Source detail: docs/module-authoring.md

28 Fly Now and Clear

Fly Now deploys one saved module into the configured Saved Games DCS instance.



Actual Windows application capture - sample data.

Before overwrite, original bytes are retained in FlyNowBackups under the user-data directory. Back up game settings separately. Inspect the installed result in DCS.

1. Deploy saved state

Close the editor after Save/Discard. Select the module, verify the destination and use Fly Now. HEIMDALL validates the payload; DCS is not launched.

2. Understand the active lock

Only one module can be active. The original destination and copied-file manifest persist. Clear before switching module or destination.

3. Clear tracked files

Clear removes the exact tracked payload, leaving module authoring data and unrelated files. It does not automatically restore the files that existed before deployment.

Source detail: docs/fly-now.md

29 Prevent inherited axis conflicts

An empty axis diff does not necessarily clear a game’s default axes. Cleanup needs exact aircraft commands and controller identities.

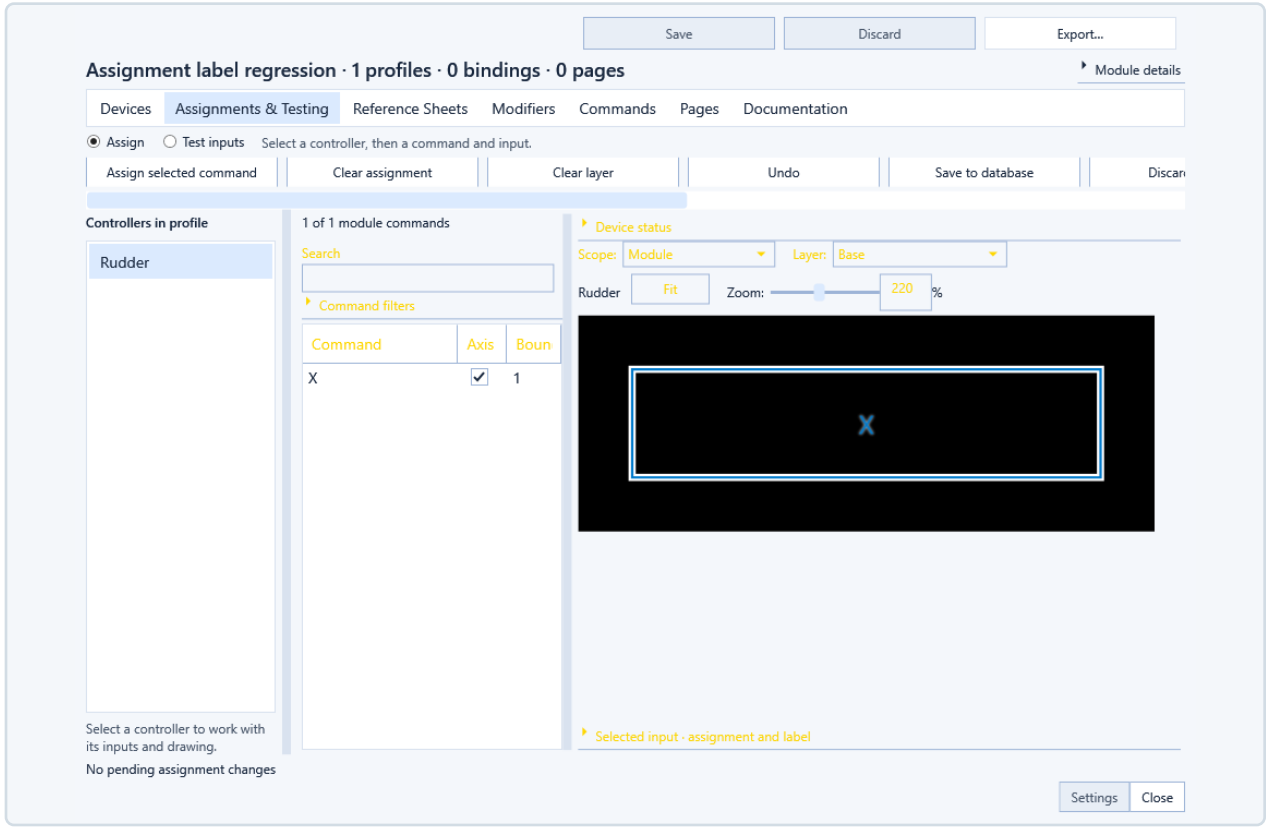
Symptom / task	What to review
Unexpected pitch, roll or thrust	Check DCS Axis Commands across every attached controller, including unused panels/quadrants.
Cleanup coverage	Refresh devices, save captured native identities and load the complete aircraft command catalog before rebuilding.
Need explicit default evidence	Back up controls; in DCS reset that device’s axes to defaults, clear all axes, save its profile, then restore your controls. Import that cleared profile as axis defaults.
Wrong aircraft or scope	Review the destination manually: a .diff.lua file does not prove aircraft identity. Aircraft and UI baselines are independent.
New controller attached later	Refresh/rebuild; an unobserved unit or an unknown command cannot be covered.

HEIMDALL retains active assigned combinations and tuning while emitting supported removals. Verify real DCS behavior after deployment.

Source detail: docs/axis-cleanup.md

30 Generic controllers and custom drawings

Premium/Owner can author supported inventory controls without a bundled picture, then adopt a reviewed drawing.



Actual Windows application capture - sample data.

1. Start without artwork

Adopt the unit's real button/axis/POV capabilities. Add it to the module as generic controls; supported native inputs remain assignable.

2. Import a drawing revision

Hardware drawings can export a starter draw.io file. Preserve explicit stable control IDs/native input keys, import the supported drawing subset and review its preview.

3. Adopt separately

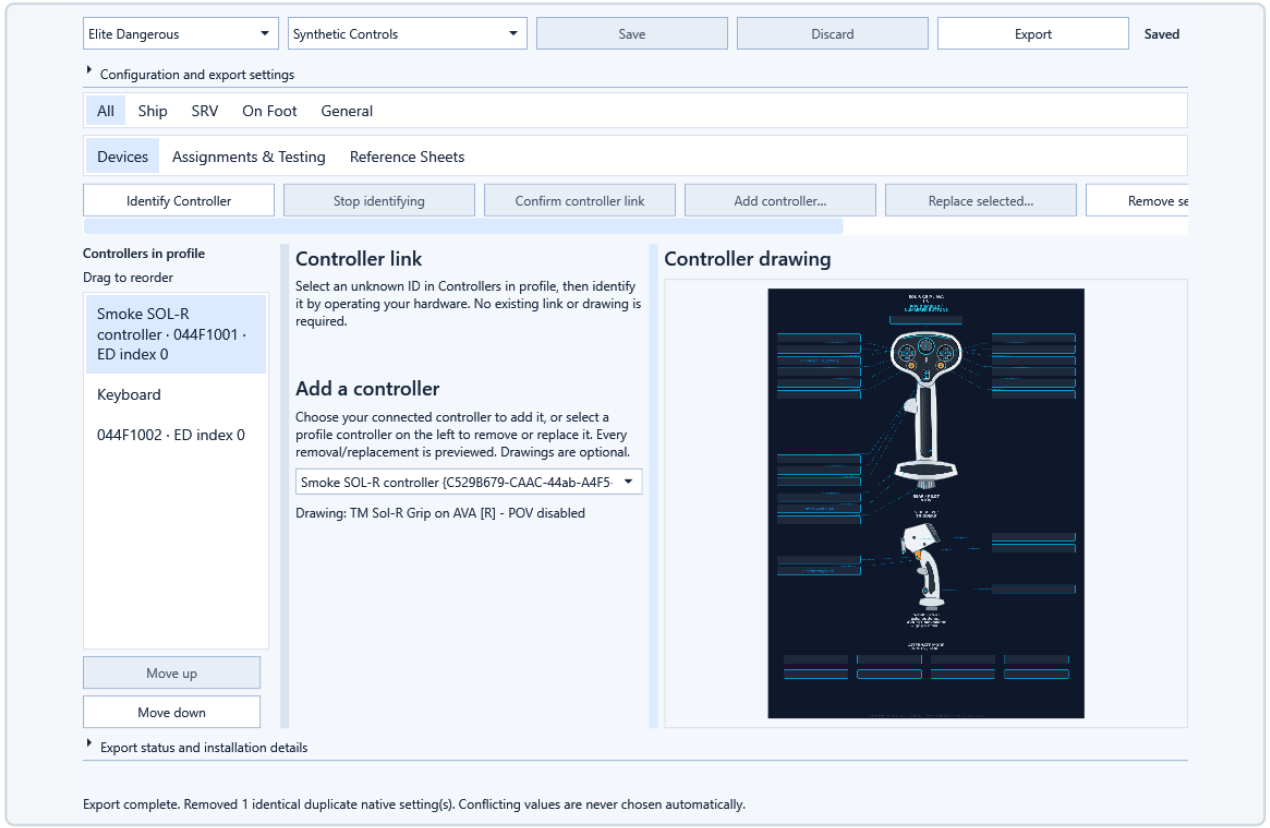
A library import creates an immutable revision. Applying it to a module or reviewing replacements is a separate staged action; compatibility checks protect existing assignments.

New artwork does not remap a command to a different input. Snapshots retain pinned drawings; deleting a library copy does not erase existing module pins.

Source detail: docs/custom-mappings.md

31 ED: import and manage configurations

ED uses named controls presets, not DCS aircraft modules or DCS modifier layers. Premium/Owner is required.



Actual Windows application capture - sample data.

1. Import a native template

Use Add / Import and select an existing working ED 4.x .binds file. Captured sources supply commands, native structure and game settings.

2. Choose a variant

New configuration clears authored bindings while retaining the template’s command definitions, controller references and settings. Duplicate preserves the current authored content in an independent copy.

3. Rename and delete deliberately

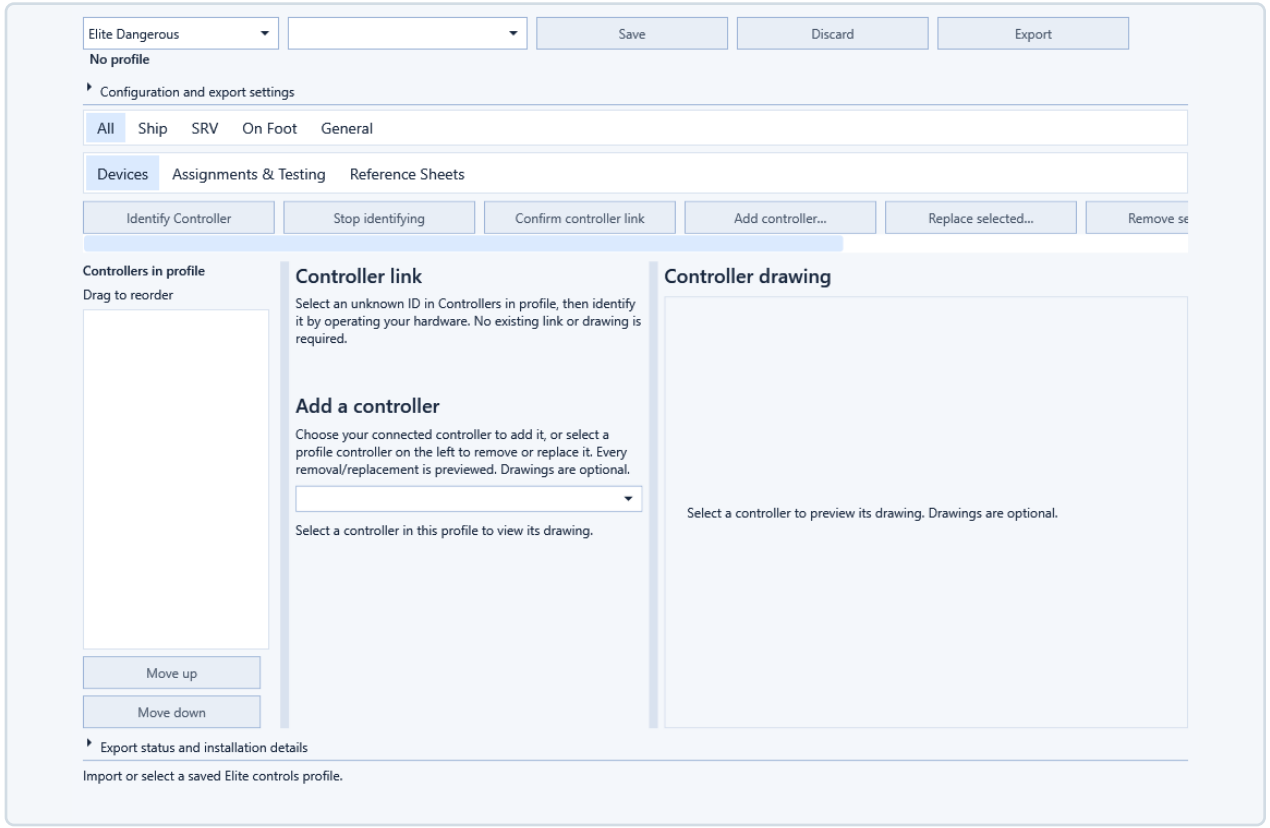
Rename stages a unique safe name; Save persists it. Delete removes the HEIMDALL configuration after confirmation, not files previously exported to ED.

HEIMDALL does not invent a game-version command catalog. New, duplicate, rename and delete resolve pending edits through Save/Discard/Cancel.

Source detail: docs/design/game-workspaces.md

32 Link a native reference to a real unit

A token such as 044F1001 identifies ED’s native device reference, not a readable controller name or your physical instance.



Actual Windows application capture - sample data.

1. Select the ED reference

In Devices, choose the target under Controllers in profile. Identify Controller listens for deliberate activity across eligible connected controllers.

2. Operate one controller

Press a button, move a hat or move an axis significantly. Initial held states establish the baseline; activity must identify a fresh, matching physical unit.

3. Confirm explicitly

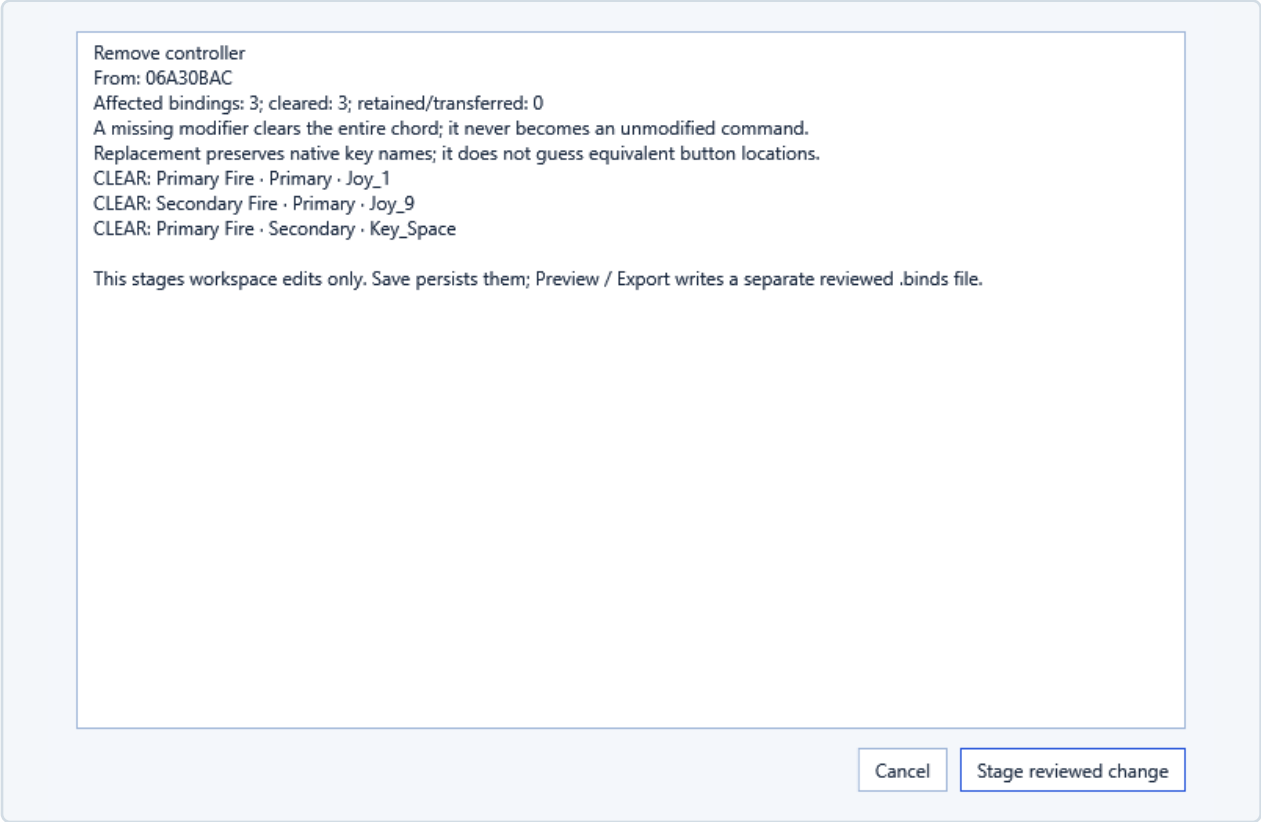
Check native product, physical GUID and captured ED index. Confirm controller link stages the association; Save persists it. Identification alone does not change assignments.

With identical products, activity identifies the physical unit but cannot prove ED’s DeviceIndex. Capture indexed references in ED and select the intended index; Windows connection order is not a substitute.

Source detail: docs/design/game-workspaces.md

33 Add, replace and remove controllers

These are reviewed edits to native binding references, not driver installation or automatic capability conversion.



Actual Windows application capture - sample data.

1. Add a unique supported unit

Choose a connected Windows controller and Add controller. HEIMDALL reuses an unambiguous compatible reference or adds a supported numeric reference.

2. Review replacement

Select the source reference and replacement. Capability probing checks native keys. Review retained/transferred and cleared bindings before applying.

3. Review removal

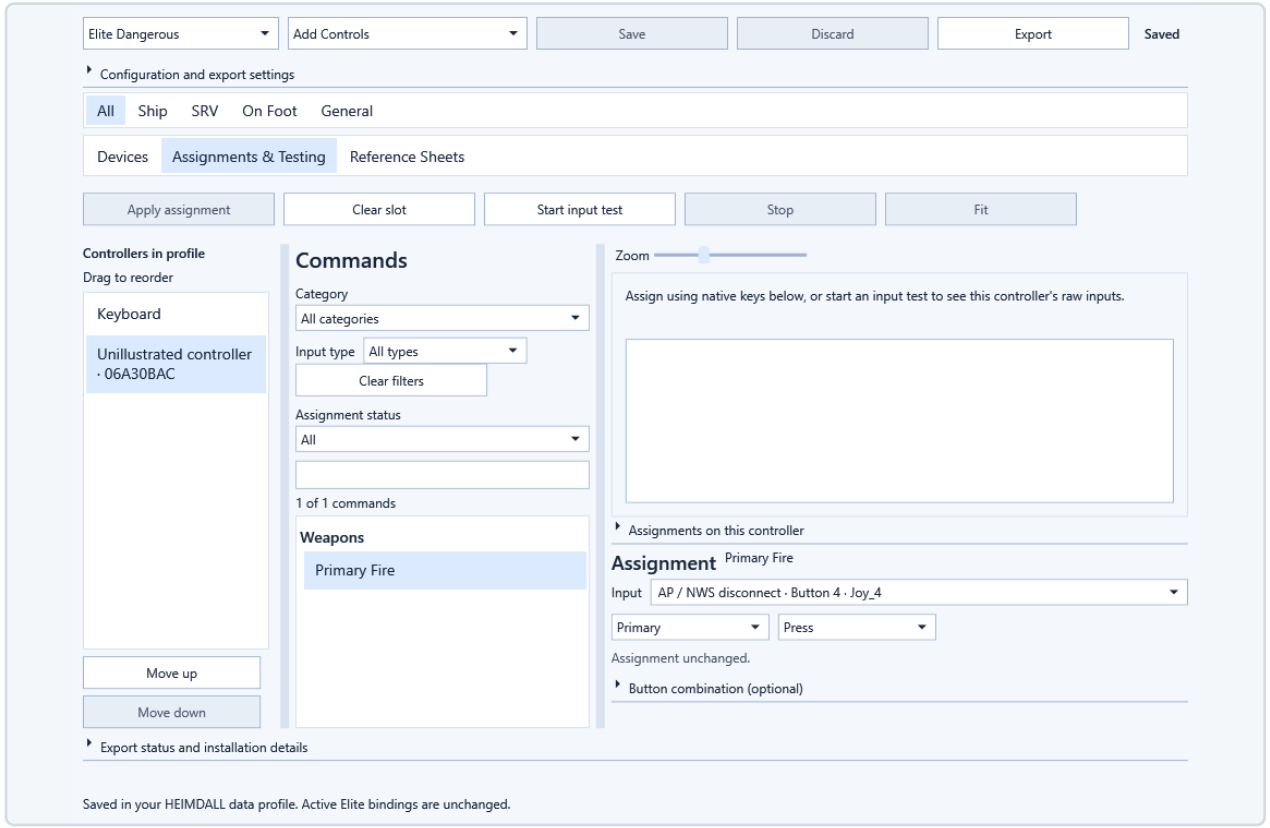
Remove selected previews affected buttons, axes and dependent combinations. A removed modifier clears its entire chord. Keyboard and mouse references are protected.

An empty added controller appears in HEIMDALL immediately, but enters native .binds output only when a binding or modifier references it. Unknown native references can block destructive edits/export.

Source detail: docs/design/game-workspaces.md

34 Organize controllers and choose a drawing

Controller order belongs to the HEIMDALL configuration and does not change native identities or ED indices.



Actual Windows application capture - sample data.

1. Reorder for your workflow

Drag above/below another entry, use Move up/down, or Ctrl+Up/Ctrl+Down. Save persists display order; Discard restores saved order.

2. Choose artwork independently

An exact unique drawing can be selected automatically. Change drawing lets you choose another compatible illustration or No drawing.

3. Work without a picture

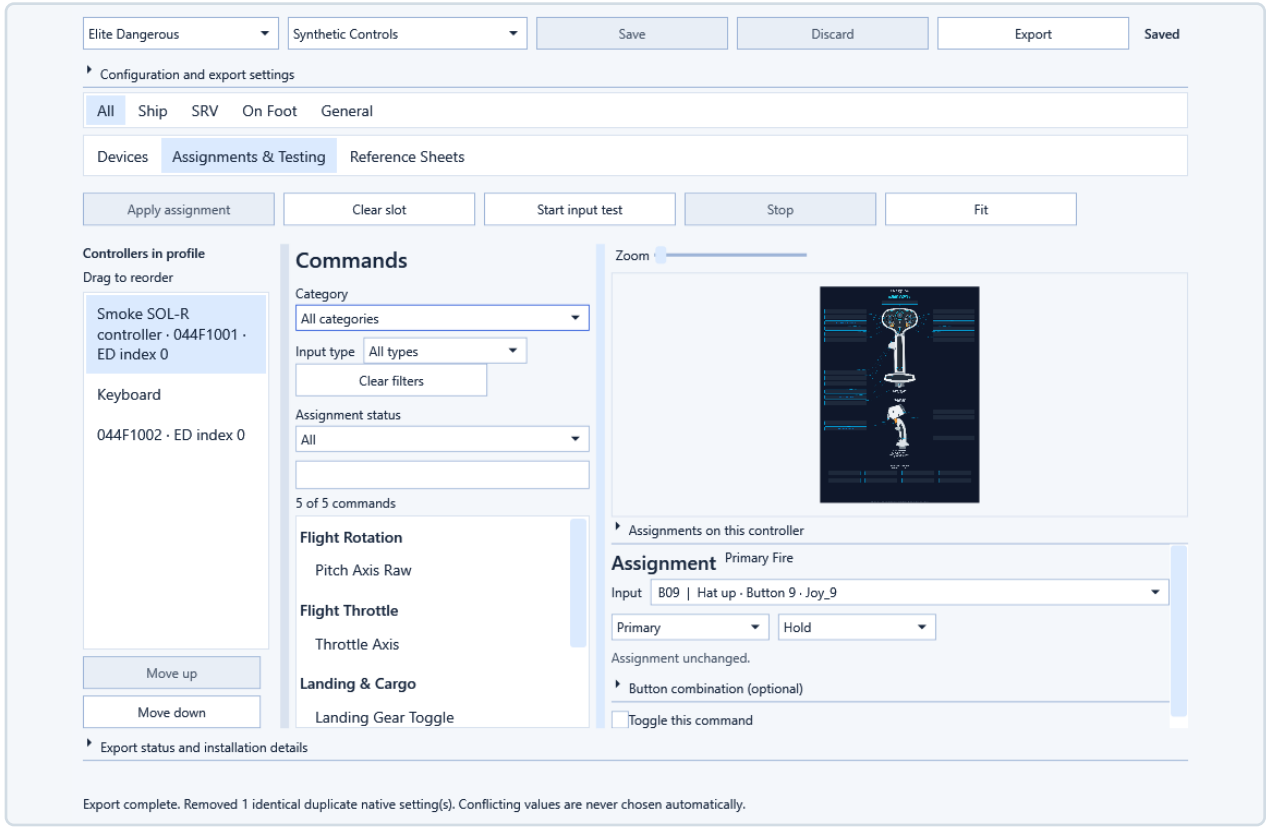
Native input choices and the assignment table remain available. A drawing name does not establish product identity or prove an attached grip.

Keep controller linking, native reference/index, and illustration as three separate decisions. A label or picture must not be used to guess an ED device ID.

Source detail: docs/design/game-workspaces.md

35 Find commands without losing context

Sections, category, type, assignment status and search intersect. Filters organize the palette without changing saved bindings.



Actual Windows application capture - sample data.

1. Choose the activity

Select All, Ship, SRV, On Foot or General. Then narrow Category and Input type to the task.

2. Search precisely

Search matches display names, native IDs and categories. Unknown command families remain accessible through Other or General.

3. Reveal an existing assignment

Click a bound control or assignment row. The editor loads its current command and settings; conflicting command filters are cleared so the command can be seen.

The selected controller, entire-profile assignment status and currently displayed section are different scopes. Reference exports use their own section/content options.

Source detail: docs/design/game-workspaces.md

36 The five assignment-status filters

Use controller-local filters to finish one device, and whole-profile filters to find global gaps.

Filter	Meaning
All	Every command matching the other filters.
Assigned Only	Has a primary, secondary or axis binding on the selected controller.
Not Assigned	No binding on this controller; it may be assigned elsewhere.
Not Assigned by Any	No binding anywhere in the complete profile.
Assigned by Any	A binding exists anywhere, including keyboard/mouse.

Count owners, not modifier references

A device used only as a combination modifier does not own the command.

Review the whole profile

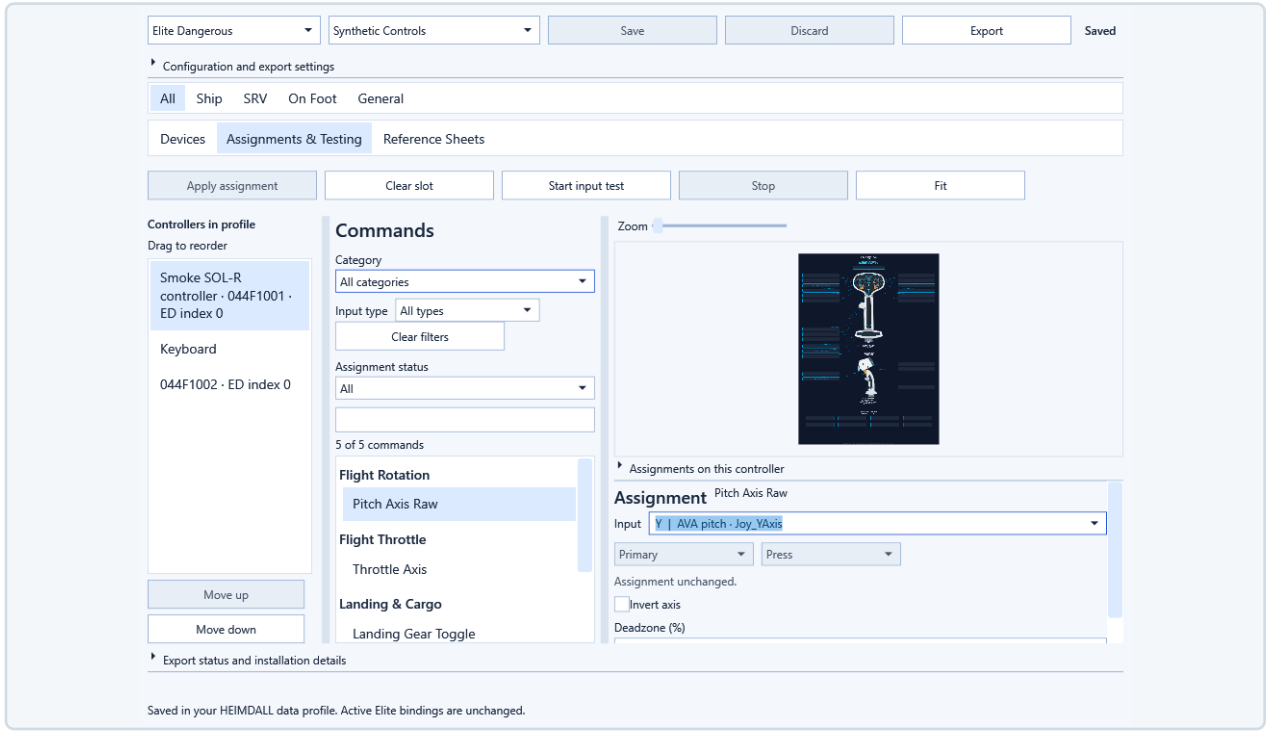
Whole-profile ownership is evaluated before the current section/category/type/search intersection. Clear filters returns the status choice to All.

Use Not Assigned by Any for genuine profile gaps. Not Assigned is useful when designing an additional controller layout.

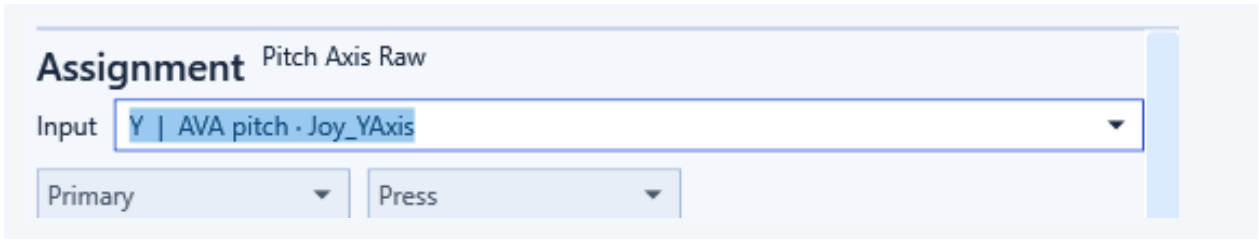
Source detail: docs/design/game-workspaces.md

37 Apply the selected assignment

Select the controller, command, compatible native input and slot. Review the loaded binding before replacing it.



Actual Windows application capture - sample data.



1. Use native input choices

The editable input list follows the selected device and command type. Entries show the default caption, explicit button number where available, and exact native key.

2. Keep the slot explicit

Buttons use Primary or Secondary. Axes use their native axis binding. Existing bindings load their values into the inspector when selected.

3. Apply and save

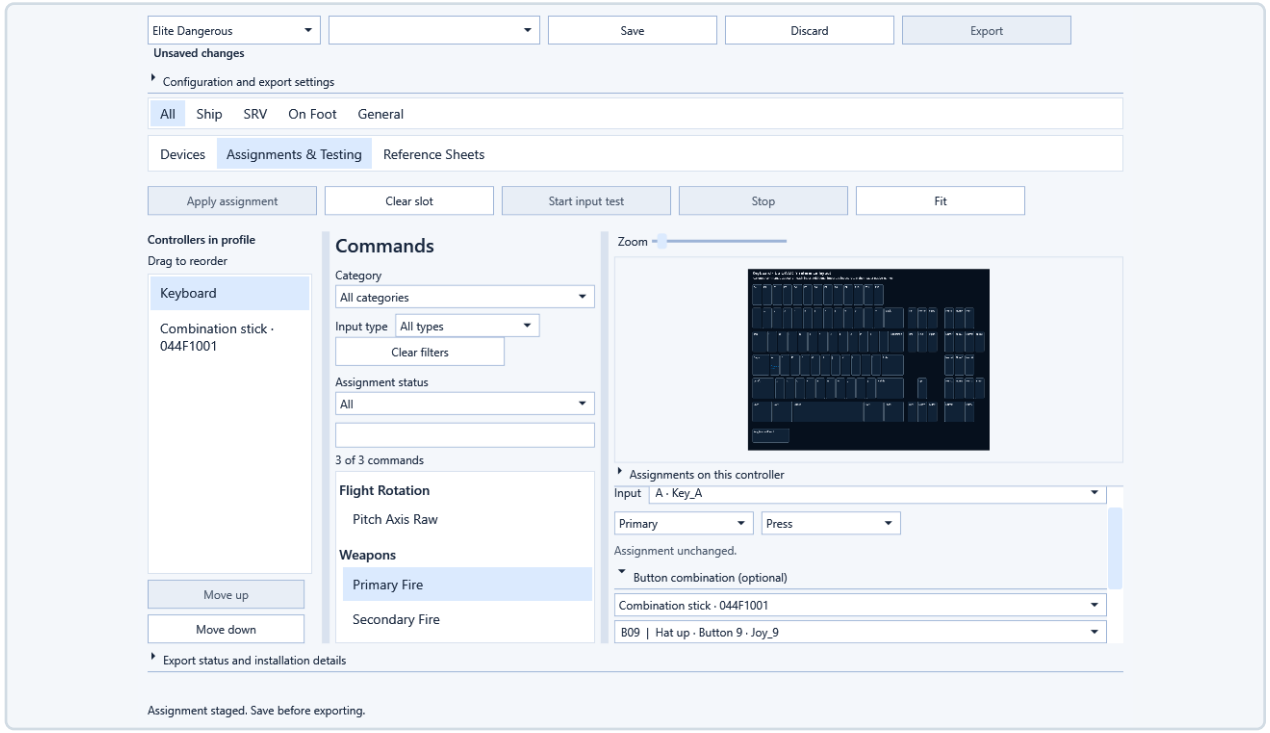
Apply assignment stages the reviewed draft. Clear slot removes that slot. Save persists the configuration; Export writes the native preset later.

> Without a drawing, captured native inputs remain listed and exact manual native-key entry remains available. A display caption is never exported in place of the native key.

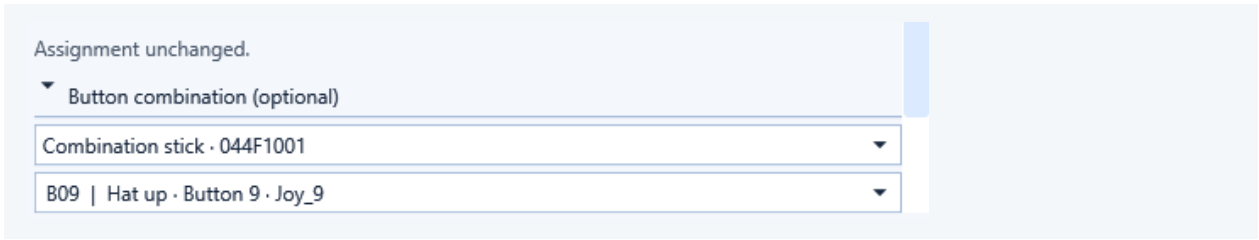
Source detail: docs/design/game-workspaces.md

38 Button combinations, Hold and Toggle

ED supports native per-binding modifier requirements. They are not DCS global Hold/Toggle modifier layers.



Actual Windows application capture - sample data.



1. Add a combination

Expand Button combination (optional). Choose the modifier's device and native input; cross-device buttons/keys are supported.

2. Choose binding Hold deliberately

Press/Hold applies to that binding. A command's native Toggle option is separate and is shown only when applicable.

3. Read the resulting command

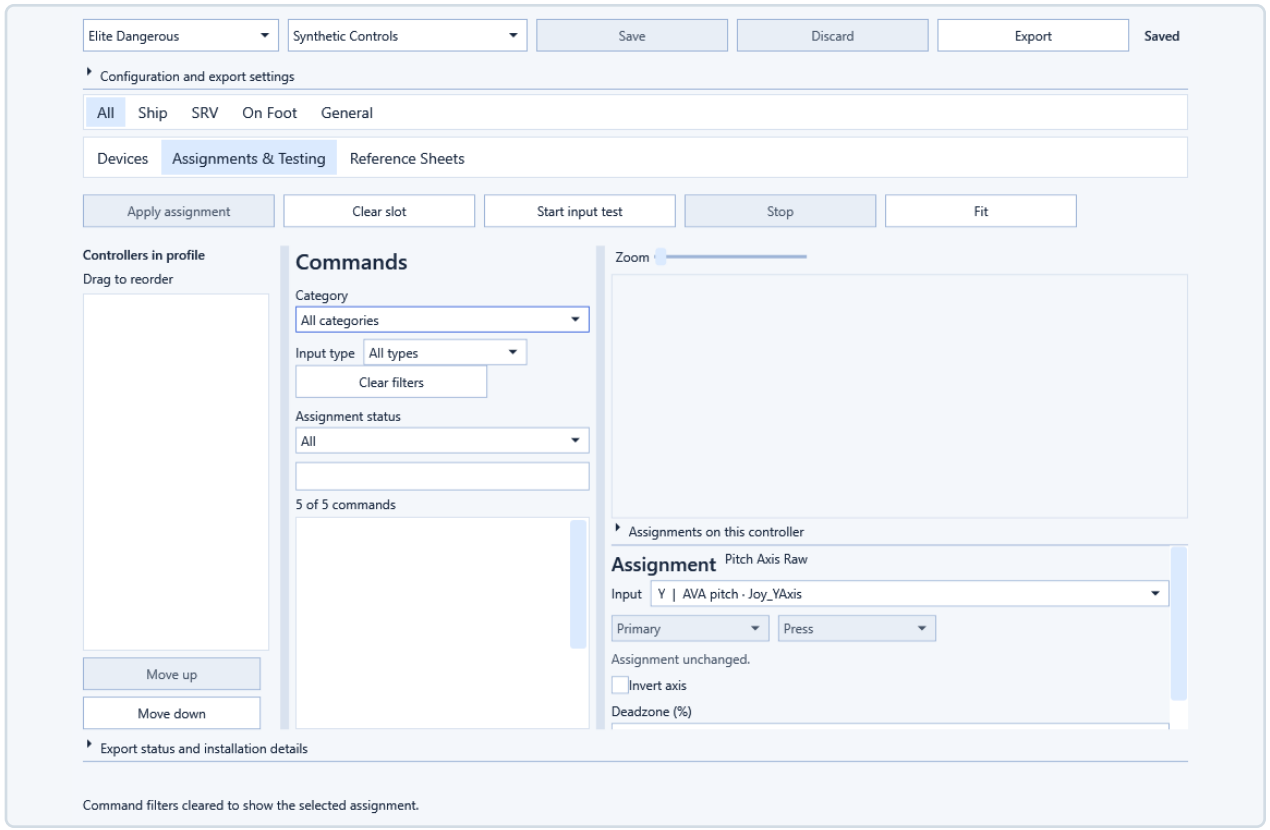
Artwork shows assigned command names rather than a raw modifier chord layer. Assignment/reference tables retain the full combination, slot and Hold/Toggle details.

Changing combination source clears its pending input choice. Export preserves native modifier device/index references rather than the displayed device captions.

Source detail: docs/design/game-workspaces.md

39 Axes and live input feedback

An axis assignment has a native key, inversion and percentage deadzone. Raw feedback tests the physical input, not the game command.



Actual Windows application capture - sample data.

1. Assign the correct native axis

Choose an axis command and a captured/supported native axis from that device. Verify the actual physical movement before choosing invert.

2. Start input test

A connected, explicitly linked physical instance is required. Apply/Clear wait while testing; Stop remains available. An illustration is not required.

3. Handle lifecycle changes

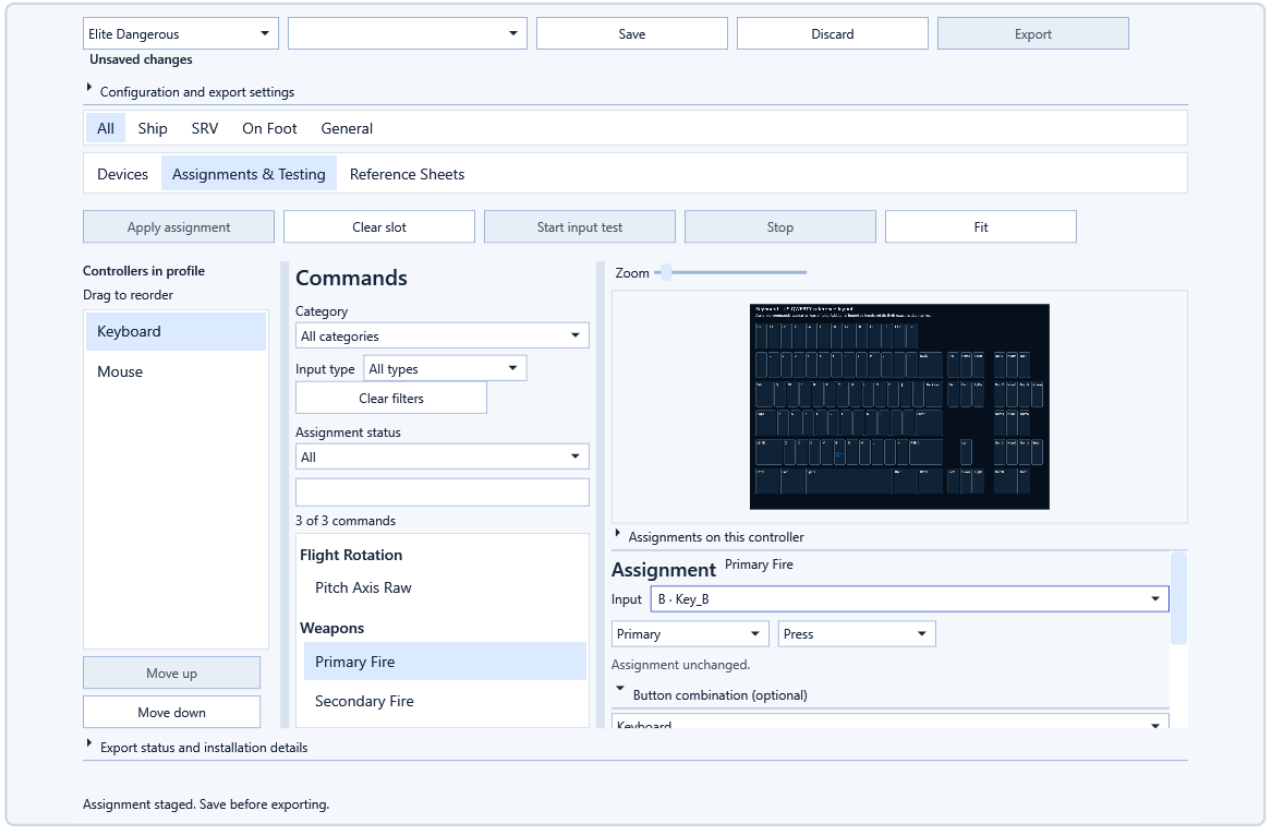
Stop, focus loss, navigation, disconnect or read failure releases input. Restart after changing the test context. Without artwork, read native activity in the text area.

HEIMDALL does not emulate Elite command execution, training prompts or hold timing. Load the exported preset in ED and test the real game behavior.

Source detail: docs/design/game-workspaces.md

40 Keyboard and mouse are first-class references

Built-in vector drawings make captured keyboard and mouse bindings visible without selecting a controller illustration.



Actual Windows application capture - sample data.

1. Select a system entry

Choose native Keyboard or Mouse. No Windows controller discovery or physical pin is required for their reference drawings.

2. Select a control

Click a key/button or assignment row. The command inspector follows it. Drag/drop and context assignment actions use the same native binding path.

3. Read every assignment

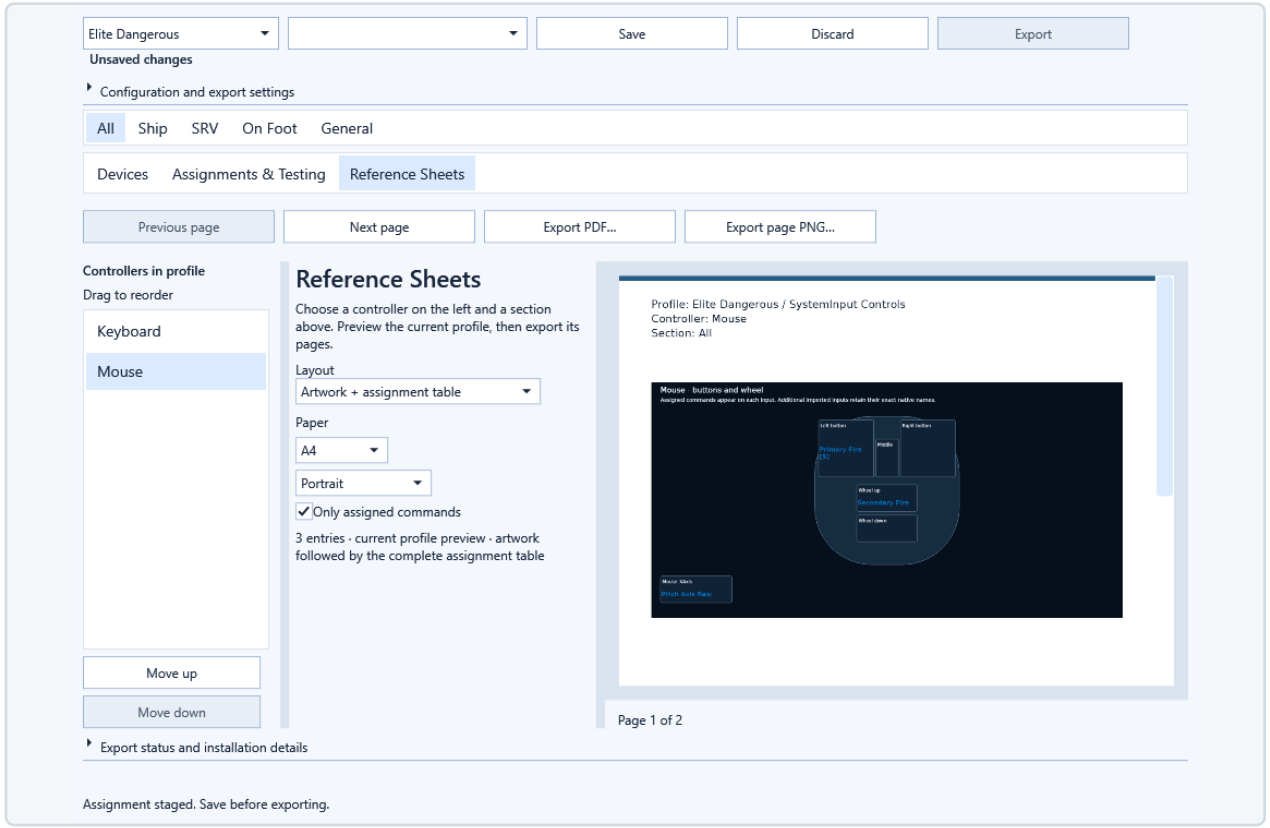
Keys/buttons show their native caption and assigned commands. Extra captured inputs appear in an additional area and in the complete table.

The keyboard is a generic US QWERTY reference layout, not automatic detection of your physical or regional keyboard. System drawings do not enable DirectInput acquisition or removable system references.

Source detail: docs/design/game-workspaces.md

41 Mouse controls and additional native inputs

Mouse references use the same assignment and reference framework as other ED devices.



Actual Windows application capture - sample data.

1. Use exact captured inputs

Built-in artwork includes left, right and middle buttons and native positive/negative wheel inputs. Captured extra buttons or mouse axes remain available as exact-name controls.

2. Keep the complete table

Unillustrated inputs and full combinations appear in the assignment/reference tables. A missing callout is not permission to invent a joystick alias.

3. Choose a practical reference

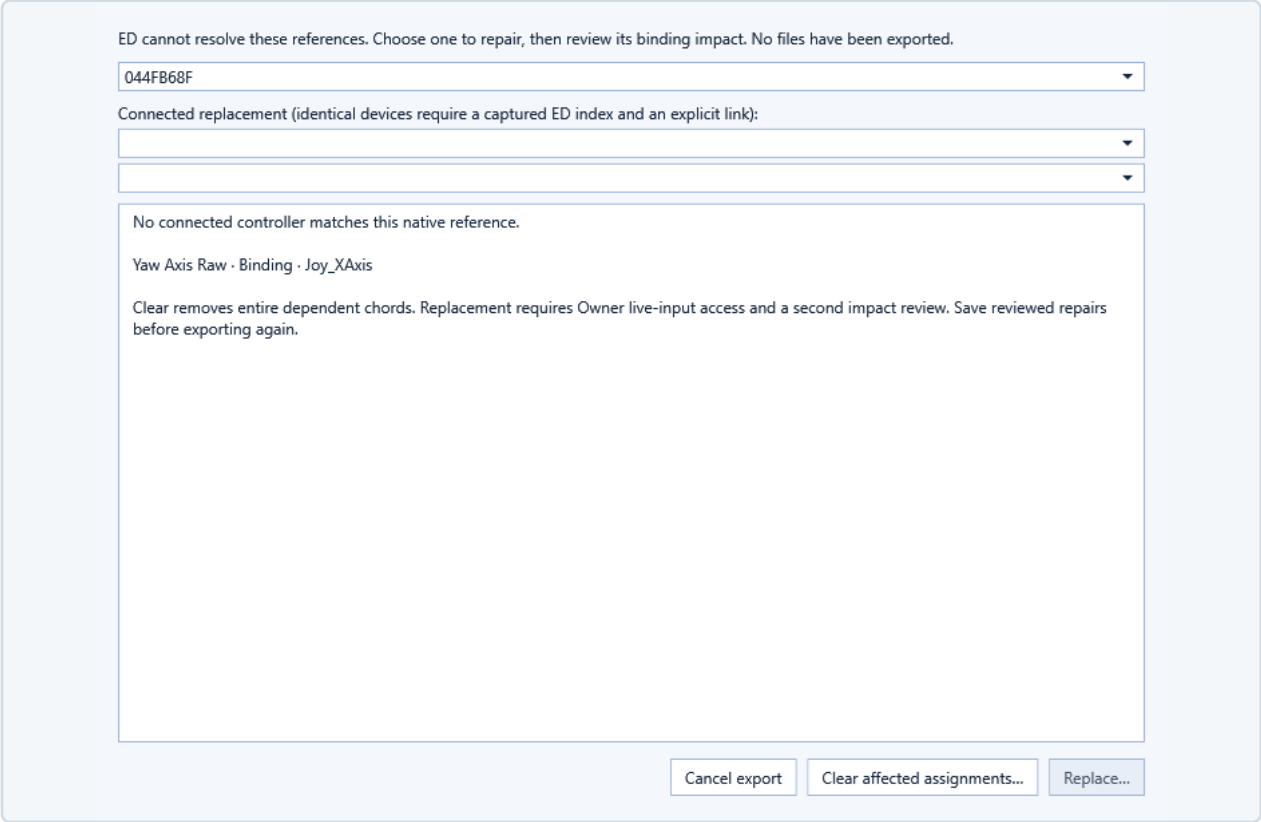
Use artwork plus tables for visual location, or a table-only layout for dense commands. Review all pages before exporting the complete controller PDF.

Keyboard and mouse are protected native system references. Their artwork does not make them physical DirectInput controllers or permit removing them from the native profile.

Source detail: docs/design/game-workspaces.md

42 Export a correctly named preset

The saved HEIMDALL configuration name becomes both PresetName and the exported native basename.



Actual Windows application capture - sample data.

1. Save and preflight

Save pending edits, close ED and use Export. Resolve missing/ambiguous devices or reviewed repair actions; save repaired drafts before trying again.

2. Choose the destination

Default: %LOCALAPPDATA%\Frontier Developments\Elite Dangerous\Options\Bindings. Configuration and export settings can remember an explicit folder override.

3. Review replacement

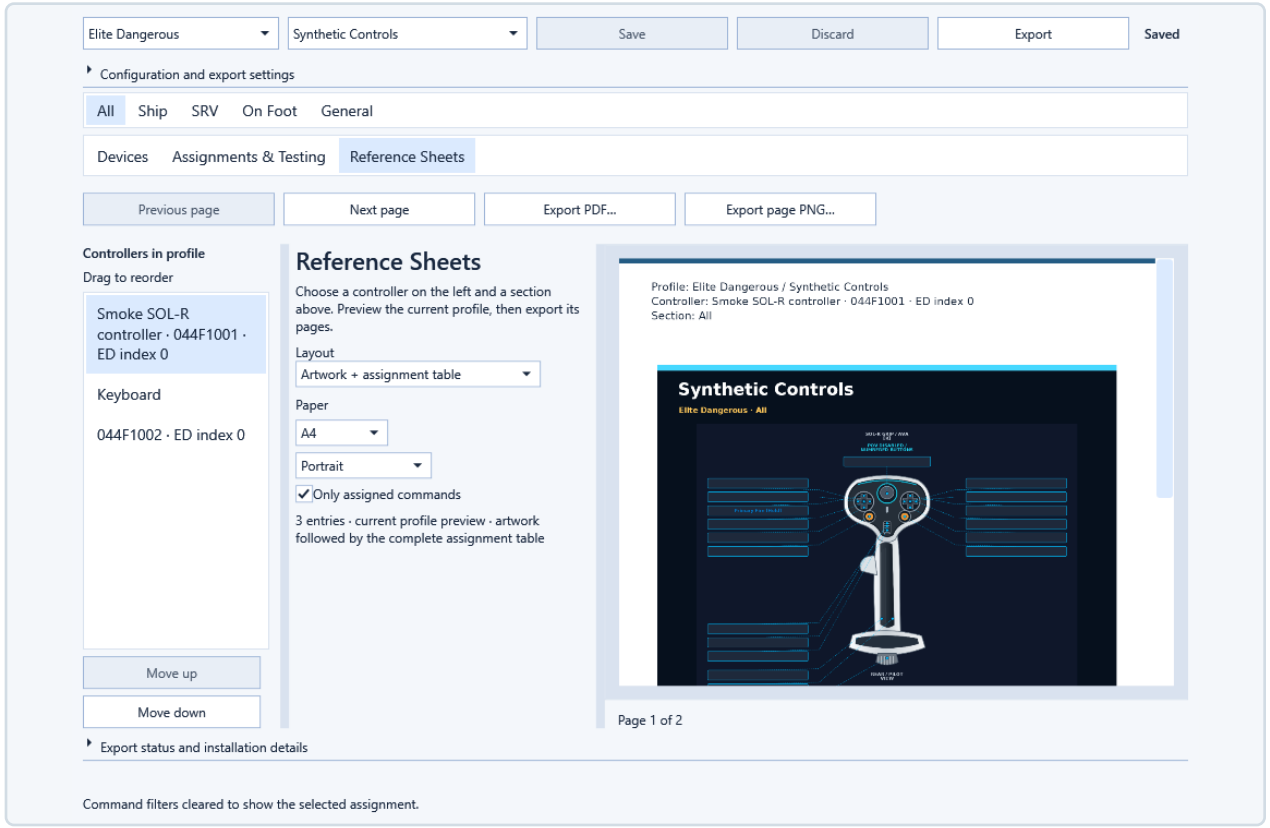
An existing same-named preset requires confirmation and an exact backup. Restart ED and select the named preset manually; .start selectors are not automatically changed.

Different safe preset names can coexist. Export to a separate review folder creates an isolated copy outside active Bindings. Renaming/deleting in HEIMDALL does not remove older exported presets.

Source detail: docs/design/game-workspaces.md

43 Reference Sheets: artwork plus complete tables

Select a controller and section, then choose artwork/table layout, A4 or Letter, and portrait or landscape.



Actual Windows application capture - sample data.

1. Choose the content

Only assigned commands is on by default. Turn it off to include captured commands unassigned on this controller.

2. Review every page

Artwork includes a complete assignment-table appendix. Devices without visible artwork fall back to tables. Long rows continue instead of being truncated.

3. Export the right scope

Export PDF saves every page for this controller. Export page PNG saves only the currently previewed page. Paper/table output is opaque white for printing.

These are portable reference documents. HEIMDALL does not install them into a native ED kneeboard folder. Print, use another display, or use an independently configured compatible viewer.

Source detail: docs/design/game-workspaces.md

44 When ED refuses or misreads a preset

Diagnose the exported native file and game error first; changing a drawing or display label cannot repair native identity.

Symptom	Check
Named preset does not appear	Verify the exported .binds destination, safe filename, XML PresetName/version, successful export status, then restart ED.
Missing device / failed GUID lookup	Read the actual ED binding-loading error. Compare native tokens, real device evidence and captured indices; do not invent aliases or substitute a product ID for a physical instance.
Two identical devices act as one	Create/capture indexed references in ED and explicitly link each instance. Do not infer DeviceIndex from Windows scan order.
No controller illustration	Choose a compatible drawing or use native inputs and tables. Artwork is optional for authoring.
Axis moves the wrong way	Check command, native axis and inversion in ED. Raw input movement alone does not prove the game’s expected direction.
Changed configuration did not reach ED	Apply, Save and Export are separate. Verify the newly exported named preset is selected in the game.

Source detail: docs/design/game-workspaces.md

45 Add another simulator without rebuilding the shell

The shared foundation supports expansion, but HEIMDALL does not currently provide an end-user Add Game plug-in wizard.

Reuse	Implement and verify for each game
Controller inventory and drawings	Native device references, identical-device indices and explicit linking rules
Shared feature/action layout	Game contexts, command catalog, supported binding modes and editable settings
Apply / Save / Discard pattern	Versioned native import/export and unknown-setting preservation
Input and reference surfaces	Axis conventions, modifier semantics and reference content
Edition policy	Every non-DCS integration requires Premium/Owner; future adapters must use the central boundary
Tests and packaging	Native fixtures, error paths, round trips, Windows smoke and in-game acceptance

Reusing a drawing does not translate commands between games. A new integration requires development and native-format evidence; do not advertise candidate games as supported.

Source detail: docs/design/multi-sim-profiles.md

46 The chapter template for the next game

Add one self-contained game chapter while leaving shared HEIMDALL fundamentals in the common section.

Chapter topic	Required evidence
Supported game/release	Native file versions, edition availability and tested limitations
Get started	Working native preset/template and exact import procedure
Controllers	Instance/product/reference/index distinction; add/remove/replace rules
Assignments	Contexts, slots, buttons, axes, combinations and toggles
Testing	Physical feedback behavior and what must still be checked in-game
Native output	Destination, naming, backups, replacement and game activation
References	Layouts, filters, print/PDF/PNG behavior
Troubleshooting	Real error messages, recovery steps and known unsupported cases

The docs/user-guide source folder keeps chapters and screenshot references modular. Replace the template with verified behavior and real captures when an integration is implemented.

Source detail: docs/design/multi-sim-profiles.md

47 Troubleshooting HEIMDALL

Start with the visible status, exact operation and the saved/draft boundary. Keep the original evidence.

Problem	Practical next step
Apply or Export is disabled	Select a valid device/input/command, stop testing, resolve validation, and Save/Discard where required. Check edition access.
Unexpected controller	Compare the exact physical GUID and native game reference. Rescan after connection changes; avoid name-based guesses.
Drawing is hard to read	Fit, enlarge the window, drag the dividers, then zoom. Layout reset is separate from deleting profile data.
Unexpected axis in DCS	Review inherited defaults and cleanup coverage, then verify DCS Axis Commands.
Restore blocks deployment	Review the restored deployment record. Forgetting bookkeeping does not remove files from Saved Games.
Need support evidence	Use F1/offline help, visible Diagnostics and support-report tools. Error logs normally live under %LOCALAPPDATA%\HEIMDALL\Logs.

A green automated test suite verifies its assertions and fixture behavior. It does not certify every controller driver, VM device mapping, monitor scale or cockpit response.

Source detail: docs/logging.md

48 Quick glossary and file map

Keep authoring data, device identity and installable output separate.

Term / file	Meaning
Module / configuration	DCS aircraft workspace / independent ED controls workspace
Physical instance GUID	A particular Windows controller instance; changes require explicit review/rebinding
Product token / DeviceIndex	ED’s native device/model reference and captured disambiguating index
Hardware template / drawing	Control definition or illustration; not a physical-device entitlement
Draft / saved state	Pending HEIMDALL work / persisted HEIMDALL document
.diff.lua / modifiers.lua	DCS native differences and modifier definitions
.binds	ED native XML preset; not the HEIMDALL working database
.heimdall-module.json	One portable DCS module snapshot
.heimdall-backup.zip	Verified full HEIMDALL profile backup
OvGME / Complete ZIP	Deployable DCS package / distribution containing the package plus documentation and snapshots

License files are account-wide and separate from profile backups. Keep a private recovery copy.

Source detail: docs/architecture/terminology.md

49 Guide scope and update record

Documentation edition 2026.10. Prepared for HEIMDALL’s current development build after the Free/Premium contract correction.

Version boundary

The repository’s release/version.json remains 1.1.1.0. This guide includes newer main-branch capabilities after PR #314; it does not assert those features exist in an older downloaded release.

Screenshot provenance

Figures are actual WPF application renders from Windows CI, using synthetic module/controller fixtures or clearly marked training sessions. No concept mockups or generated UI images are used.

Source baseline

Production behavior: main commit f710fdd9953d888f0d4f1591f55ea87aebe0111d. The screenshot-only capture harness is tracked separately in PR #315.

Maintain the guide

Keep the JSON chapter files, screenshot assets and generator together. Update changed workflows, recapture their views, regenerate the PDF, and inspect every rendered page before publishing.

Primary references: the HEIMDALL repository documentation and native-format implementation/tests. Per-page source links point to the reviewed production baseline. Manual hardware/in-game release acceptance remains separate.

Source detail: docs/README.md